A Predictive Amplitude Law for the Vibration-Driven Slip Front in Capstan Friction

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Abstract

The classical capstan relation predicts global frictional force transmission around a wrapped flexible element, but does not determine where cycle-accumulated relative motion localizes when the contact is subjected to torsional vibration. We investigate whether such vibration produces an interior creep–grip transition in a multiply wrapped rope–capstan contact and whether the centre of that transition can be predicted from independently measured mechanical quantities.

A reduced-order crossover model was derived by combining the exponential capstan resistance profile with an independently calibrated tension-dependent preslip stiffness. The resulting law predicts logarithmic dependences of the transition centre θ^* on displacement amplitude and held-end tension. All quantities entering the prediction were calibrated independently of the measured transition positions.

A total of 84 physical runs were performed using a four-wrap braided polyester rope on a torsionally oscillated aluminium capstan. The amplitude series gave $\partial\theta^*/\partial \ln u_0 = 9.54 \pm 0.74$ rad, compared with the independently predicted 10.10 ± 1.45 rad. The held-end-tension series gave $\partial\theta^*/\partial \ln T_1 = -5.56 \pm 0.50$ rad, compared with the prediction -5.56 ± 0.62 rad. Across the eight unique amplitude–tension operating points used for the absolute-prediction test, the calibration-fixed law gave an RMSE of 0.117 rad and a median absolute error of 0.094 rad. Neither control condition produced an accepted stable front across twelve control runs. At fixed measured displacement amplitude, no systematic frequency dependence of the transition centre was resolved over 30–60 Hz.

Within the tested partial-mobilisation regime, the results therefore support a compact predictive law for the fitted centre of a vibration-driven creep–grip transition in a wrapped flexible contact.

Keywords: Capstan friction, partial slip, vibration-induced friction, contact mechanics, model selection

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Table of Contents

1.0 Introduction 5

1.1 Relation to Prior Work and Research Gap . . . 6

1.2 Contributions of the Present Work . . . 7

2.0 Theoretical Framework and Model Derivation 7

2.1 Physical Picture and Model Assumptions . . . 8

2.2 Capstan Balance and Effective Resistance Profile . . . 8

2.3 Vibration-Induced Tangential Loading . . . 10

2.4 Derivation of the Transition-Position Law . . . 10

2.5 Physical Regimes and Testable Predictions . . . 11

3.0 Experimental Methodology and Data Analysis 12

3.1 Experimental Strategy . . . 12

3.2 Experimental Apparatus and Optical Measurement . . . 13

3.3 Independent Calibration . . . 14

3.3.1 Geometry and Drive Calibration . . . 14

3.3.2 Boundary Loading and Friction Calibration . . . 15

3.3.3 Tangential Contact Stiffness . . . 15

3.3.4 Optical Measurement Floors . . . 16

3.4 Confirmatory Experimental Design . . . 17

3.4.1 Amplitude and Holding-Tension Sweeps . . . 17

3.4.2 Control and Frequency Tests . . . 18

3.5 Image Processing and Creep-Rate Extraction . . . 19

3.6 Transition Identification . . . 20

3.6.1 Model Selection and Front Criteria . . . 22

3.6.2 Stability of the Transition Estimate . . . 23

3.7 Validation of the Predictive Law and Uncertainty Analysis . . . 24

3.7.1 Amplitude and Tension Dependence . . . 24

3.7.2 Absolute Prediction Test . . . 25

3.7.3 Frequency Falsification Test . . . 25

3.7.4 Uncertainty Propagation 25

4.0 Results 26

4.1 Independent Calibration Results 26

4.2 Observation and Classification of the Creep-Grip Transition 26

4.3 Amplitude Dependence of the Transition Position 28

4.4 Holding-Tension Dependence 28

4.5 Absolute Prediction of the Transition Position 30

4.6 Frequency Falsification Test 31

4.7 Uncertainty and Robustness of the Measured Transition 32

4.8 Condition-Level Experimental Summary 33

5.0 Discussion 33

6.0 Evaluation and Error Analysis 35

7.0 Conclusion 37

References 39

A Experimental Operating Conditions and Transition Measurements 40

B Experimental Parameters, Calibration, and Reproducibility 41

C Transition Classification and Robustness Diagnostics 43

1.0 Introduction

A rope wrapped around a cylindrical surface can restrain a load substantially greater than the force applied at its held end. The classical capstan effect describes the exponential amplification of transmissible tension with friction and cumulative wrap angle (1). Tangential vibration is also known to modify the frictional response of contacting bodies (2, 3). These two phenomena raise the question of how the two may relate to one another: when a rope that is otherwise statically wrapped around an object experiences torsional vibration, does cycle-accumulated relative motion develop throughout the entire wrapped contact, or only over part of the wrap?

Our hypothesis is that relative motion will initiate at the end of the rope that is applied with the least tension, and continue towards the loaded end of the rope as the amplitude of the vibrations increases. We therefore expect a region along the wrap where relative motion between the rope and cylinder accumulates over time (creeping region), and another region where the relative motion does not accumulate over time (gripping region). These regions are separated by a transition centred at θ^*, measured from the held-end contact point. Figure 1 illustrates this hypothesised arrangement, both on the wrapped contact and on the unwrapped cumulative coordinate $0 \leq \theta \leq 8\pi$ used throughout this work.

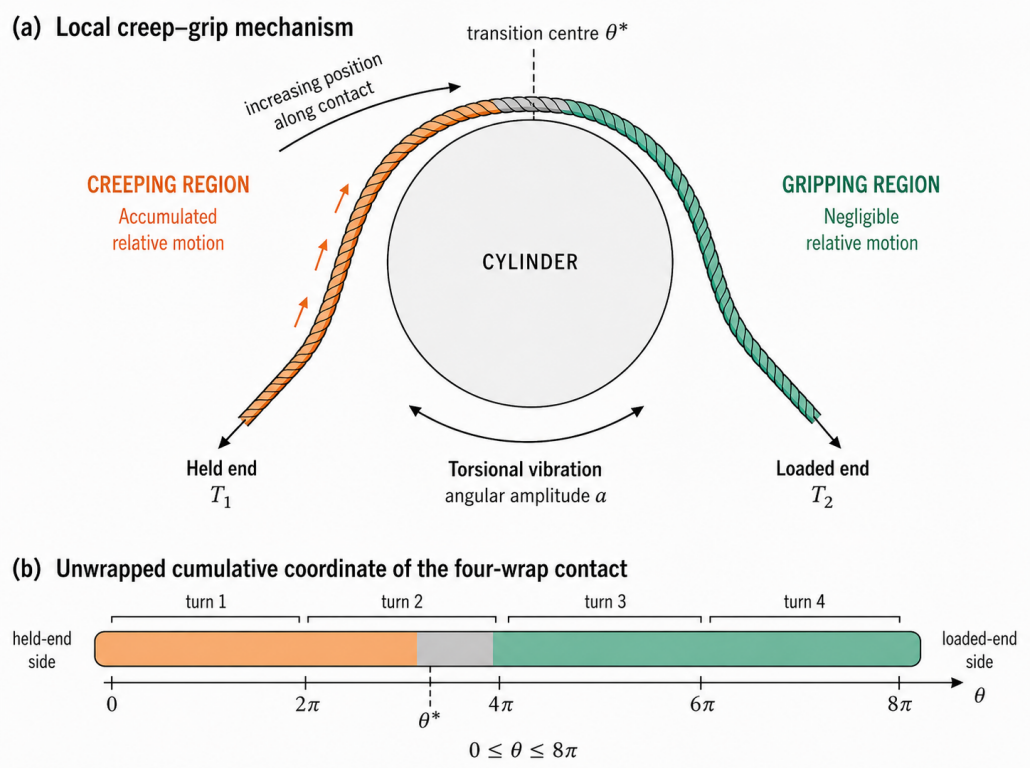


Figure 1: Conceptual representation of the hypothesised vibration-driven creep-grip transition. **(a)** Local mechanism: torsional oscillation of the cylinder produces cycle-accumulated relative motion near the lightly tensioned held end, while the more highly tensioned region remains gripping; the transition is centred at θ^*. **(b)** Unwrapped cumulative coordinate used for the four-wrap experiment, with $0 \leq \theta \leq 8\pi$. The illustrated position of θ^* is schematic and does not correspond to a particular experimental condition.

Throughout this work, the terms *creep*, *grip*, and *front* are used operationally rather than as idealized contact states. "Creep" denotes cycle-accumulated displacement of the rope relative to the adjacent capstan surface. "Grip" denotes a region in which this accumulated relative motion remains below the independently calibrated creep-rate measurement floor; it does not imply the complete absence of reversible deformation or microslip. The transition between these responses occurs over a finite angular width, and θ^* denotes the fitted centre of this transition rather than a mathematically discontinuous boundary.

Likewise, the effective quantity $T_{\text{eff}}(\theta)$ introduced below is a reduced-order resistance variable and should not be

interpreted as the instantaneous rope tension at every phase of the vibration cycle. These definitions delimit the quantities directly tested by the experiment.

1.1 Relation to Prior Work and Research Gap

The classical capstan relation describes the exponential transmission of tension through friction over a wrapped flexible element (1, 4). Extensions of capstan and belt mechanics have considered effects including finite compliance (5), bending rigidity (6–8), creep (9, 10), and partial mobilisation of friction (11, 12). More generally, partial-slip contact mechanics establishes that sticking and slipping regions can coexist within a frictional contact rather than requiring simultaneous gross sliding throughout the interface (13–16).

Creep theory, following Reynolds (9) and Swift (17), and the shear theory of Firbank (11) consider the belt wrap to include an arc of both adhesion and slip. For the static case with no gross slip between the belt and the wheel, Lubarda (12) showed that the arc of mobilised friction between the belt and the wheel adjacent to the pull end of the belt has a subtended angle of $\theta_s = \mu^{-1} \ln(T_2/T_1)$. However, the present work includes three distinguishing features. First, the direction of the mobilisation is reversed: in existing theories, the arc of mobilised friction grows from the high-tension end towards the held end, whereas the present study shows that mobilisation occurs next to the lightly tensioned held end of the belt. Second, the position of the mobilisation boundary in the existing theories is entirely a function of the ratio of the belt tensions and the coefficient of friction between the belt and the wheel. Lubarda (12), for instance, states that the angle θ_s is independent of the elastic stiffness of the belt. Several established belt-drive creep models are formulated around a non-zero mean transport velocity between the belt and the pulley. The presented configuration posed in this study is deliberately at zero mean transport velocity, isolating the torsional vibrations as the sole mobilising agent and enabling the spatial response to be assigned to it clearly. The importance of torsional vibrations in belt drives is established by Tonoli et al. (18), who extended Firbank's model for belt drives to include small oscillations around a non-zero mean rotational speed of the pulley. In that model, the authors established that the torsional vibrations of the system are associated with the shear response of the belt drive, which can be described by a lumped model of rotational dynamics of the drive. The phenomenon explored in the present study is where along the wrap of the belt the response to torsional vibrations changes.

While these studies reveal the effect of vibration upon friction, they assume the contact is spatially uniform. The present work introduces the missing spatial dependence, allowing the transition position itself to be predicted.

A separate body of work has shown that tangential vibration can modify the effective frictional response of contacting bodies (2, 3). These studies establish that vibration can alter the conditions under which relative motion occurs, but they do not provide a spatial law for a multiply wrapped rope–capstan contact.

Hence, the unresolved question addressed here is whether, under torsional vibration of an otherwise statically loaded capstan, the *position* of an interior creep–grip transition can be predicted quantitatively from independently measured mechanical properties.

The present work therefore combines two established physical ingredients: the exponential spatial variation associated with capstan friction and the modification of tangential contact loading under vibration. From these, a reduced-order crossover model is derived for the transition centre θ^*. The central test is stronger than demonstrating that a transition exists: the model parameters are calibrated independently of the measured transition positions, after which the predicted amplitude dependence, held-end-tension dependence, absolute transition positions, onset behaviour, and frequency response are compared with experiment.

1.2 Contributions of the Present Work

The central contribution of the present work is a change in what the capstan relation is asked to predict. The classical relation predicts a single global number: the tension ratio at impending gross slip. The result established here is that the same frictional gradient that determines the tension ratio also determines a spatial coordinate. When a loading mechanism that grows with wrap angle at a rate slower than the resistance to slip is introduced, the condition for slip resistance is satisfied over a bounded spatial position; the endpoint of that interval is a measurable and predictable position.

The specific contributions of this investigation are:

1. a reduced-order closed-form law for the centre θ^* of a vibration-driven creep–grip transition in a multiply wrapped rope–capstan contact;
2. independent experimental calibration of the geometric, frictional, contact-stiffness, loading, and vibration quantities entering that law;
3. spatial optical measurement of cycle-accumulated rope–capstan relative motion over a four-wrap contact;
4. model-based discrimination between a stable localized transition and alternative responses including negligible spatial structure, distributed creep, migrating transitions, and multiple localized regions; and
5. confirmatory tests of the predicted amplitude and held-end-tension dependences, absolute transition position, onset behaviour, and absence of a resolved frequency dependence over the tested 30–60 Hz range.

Two features of the result ensure its reach beyond the system tested. First, the sensitivity $\partial\theta^*/\partial \ln T_1 = -1/\mu_{\mathrm{red}}$ to held-end tension is independent of the stiffness exponent α, and shares the same $1/\mu$ structure as the static pre-gross-slip result (12); the two results thus corroborate one another as consequences of the same exponential resistance profile. The sensitivity to amplitude, $1/[(1-\alpha)\mu_{\mathrm{red}}]$, has no counterpart in the static case, and is thus a feature of the vibration law alone. Second, the law's expression as a ratio of two independently measurable force scales suggests a route for extension to other wrapped contacts in which a gradient in frictional resistance and a preslip contact stiffness can be characterised.

The practical consequence of the law is that partial mobilisation under vibration can be made locatable. The types of wrapped frictional contact that are known to exist in engineering systems, but which operate well below their limit of gross-slip, include mooring and winch lines subjected to vibration from the machinery to which they are attached. Others, such as tensioning systems in manipulators that are actuated by vibrational means, capstan clutches and brakes, and textile processing systems, all exhibit wear along the wrap that is currently not predicted by any existing means. The law developed here, however, allows for the location of areas of wear on these systems based upon parameters that can be measured before the system is placed into service.

The capstan relation, partial-slip concepts, statistical model-selection methods, and bootstrap resampling procedures used in this work are established tools. The original contribution is their combination into, and experimental test of, a predictive spatial law for the vibration-driven transition position.

2.0 Theoretical Framework and Model Derivation

The model predicts the angular position at which the vibration-induced tangential load is equal to the local frictional resistance of the rope. This angle is denoted by θ^* and is measured from the held end of the rope wrapping around the cylindrical capstan.

The derivation of this result has three parts. First, the relation between the tension along the rope and the frictional resistance along the wrap is described. Second, the torsional displacement of the cylinder is related to the tangential contact force between the rope and the cylinder. Finally, equating these two expressions results in a closed-form expression for the angle θ^*.

2.1 Physical Picture and Model Assumptions

Consider a cylinder around which a rope is wrapped such that the total wrapping angle is Θ. Let T_1 be the tension in the rope at its held end, and let T_2 be the tension in the rope at its loaded end. The angular coordinate θ is measured from the held end of the wrap to the loaded end of the wrap. For the four-wrap configuration,

$$0 \leq \theta \leq \Theta, \qquad \Theta = 8\pi. \tag{1}$$

If there were no vibration of the cylinder, the tension in the rope would increase along the wrap due to friction alone. As a result, the frictional resistance of the rope at any given point along the wrap would be smallest at the held end of the wrap and increase along the wrap towards the loaded end.

When the cylinder undergoes torsional vibration, its surface moves tangentially relative to the rope. This repeated relative motion generates an oscillatory tangential force at the rope–cylinder contact. The local contact state is determined by the competition between vibration-induced loading and local frictional resistance.

The position of transition between the two types of loading is defined to be that which satisfies the condition of equality of these two quantities. For angles of wrap between 0 and θ^*, the loading applied to the rope is stronger than the local resistance of the wrap; for angles of wrap between θ^* and Θ, the resistance of the wrap to the movement of the rope is stronger than the loading that is applied to those wraps. Thus, the model indicates that the region of vibration influence develops at the lightly tensioned held end of the rope, rather than that the entire rope must simultaneously lose its grip on the drum.

The transition between these two states need not be discontinuous. The two states may transition over some finite angle. The angle θ^* is the centre of this angle.

The model makes three principal assumptions. First, the rope is treated as bending-flexible and its self-weight is neglected, while axial extensibility is neglected in the reduced-order description. Second, that the resistance to motion is represented by an effective coefficient of friction, which is determined from the relative motion of the rope and drum at slow rates of motion. Finally, the tangential loading that is applied to the drum is related to the displacement of the drum from its static position through the effective stiffness of the preslip contact between the rope and the drum.

2.2 Capstan Balance and Effective Resistance Profile

Consider an element of rope of infinitesimal arc length, subtended by an angle $d\theta$ at the centre of the cylinder, as shown in Fig. 2. The rope element experiences tensions $T(\theta)$ and $T(\theta + d\theta)$ at its two ends, and the cylinder exerts a normal reaction force dN and a tangential friction force dF on the element of the rope.

To first order in $\mathrm{d}\theta$, radial equilibrium of the infinitesimal element gives

$$\mathrm{d}N = 2T(\theta) \sin\left(\frac{\mathrm{d}\theta}{2}\right) \simeq T(\theta)\,\mathrm{d}\theta. \tag{2}$$

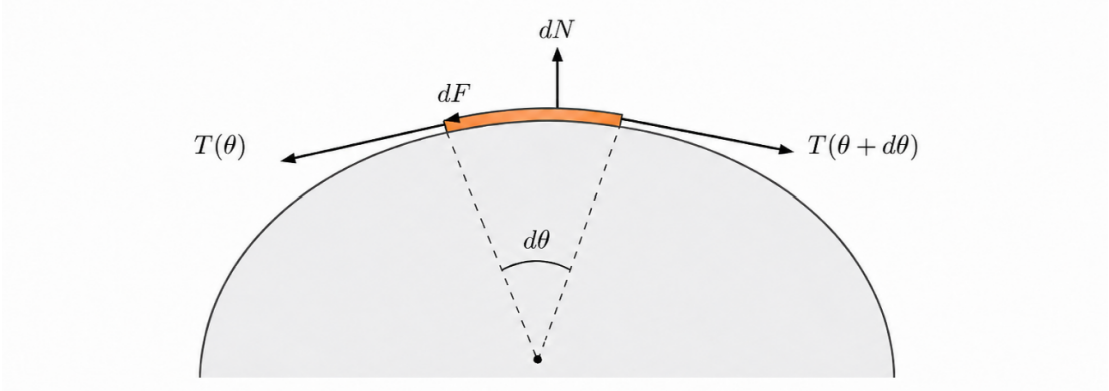


Figure 2: Free-body diagram of an infinitesimal rope element subtending an angle $\mathrm{d}\theta$ at the centre of the cylinder. The tensions at its two boundaries are $T(\theta)$ and $T(\theta + \mathrm{d}\theta)$, while $\mathrm{d}N$ and $\mathrm{d}F$ denote the normal reaction and tangential friction force exerted by the cylinder on the rope.

At a frictionally mobilised state, $\mathrm{d}F = \mu\,\mathrm{d}N$, and tangential equilibrium requires $\mathrm{d}T = \mathrm{d}F$. Therefore,

$$\frac{\mathrm{d}T}{T} = \mu\,\mathrm{d}\theta. \tag{3}$$

Integrating from the held-end boundary $T(0) = T_1$ to an arbitrary angular position θ,

$$\int_{T_1}^{T(\theta)} \frac{\mathrm{d}T'}{T'} = \int_0^{\theta} \mu\,\mathrm{d}\theta', \tag{4}$$

gives the classical capstan relation

$$T(\theta) = T_1 \exp(\mu\theta). \tag{5}$$

Equation 5 describes the limiting capstan balance when local friction is fully mobilised. The vibration experiment, however, contains coexisting creeping and gripping regions and therefore does not require the entire four-wrap contact to satisfy that limiting state. We instead retain the exponential capstan form as a reduced description of the spatial resistance scale and determine its effective coefficient independently from slow-creep calibration.

This reduced resistance profile is written as

$$T_{\mathrm{eff}}(\theta) = T_1 \exp\left(\mu_{\mathrm{red}}\theta\right), \qquad 0 \le \theta \le \Theta, \tag{6}$$

where μ_{red} is obtained independently from slow-creep calibration. The quantity $T_{\mathrm{eff}}(\theta)$ is not intended to represent the instantaneous physical rope tension at every phase of the vibration cycle. Instead, it is a reduced-order resistance-state variable: the classical capstan-form spatial dependence is retained, while its exponential growth rate is parameterised by the independently measured slow-creep coefficient μ_{red}.

The use of T_{eff} is therefore a closure for the spatial resistance profile rather than a solution of the instantaneous dynamic tension field. Its role is to provide the local resistance scale against which the imposed vibration-induced tangential loading is compared. Consequently, the model is intended to predict the centre of the experimentally observed creep-grip transition, rather than the complete time-dependent distributions of rope tension, contact pressure, microslip, or interfacial stress within the rope-capstan contact.

2.3 Vibration-Induced Tangential Loading

Torsional oscillation of the capstan imposes a tangential displacement on the rope–cylinder interface. If the measured peak angular amplitude is a, the corresponding peak surface displacement is

$$u_0 = ra, \tag{7}$$

where r is the capstan radius. The transmitted tangential force also depends on the effective preslip stiffness of the rope–aluminium contact. Its measured tension dependence is represented by

$$\kappa_t(T) = \kappa_{\mathrm{ref}} \left(\frac{T}{T_{\mathrm{ref}}} \right)^{\alpha}, \tag{8}$$

where κ_{ref} is the tangential stiffness per unit wrap angle at reference tension T_{ref}, and α is the measured load-dependence exponent.

For an infinitesimal element $\mathrm{d}\theta$, the vibration-induced tangential-force increment and local static-friction resistance are

$$\mathrm{d}Q_{\mathrm{vib}} = \kappa_t(T_{\mathrm{eff}}(\theta))\, u_0\, \mathrm{d}\theta, \tag{9}$$

$$\mathrm{d}Q_{\mathrm{res}} = \mu_s T_{\mathrm{eff}}(\theta)\, \mathrm{d}\theta, \tag{10}$$

where μ_s is the independently measured static coefficient of friction. Dividing by $\mathrm{d}\theta$ and substituting Eqs. 6–8 gives the corresponding force scales per unit wrap angle,

$$q_{\mathrm{vib}}(\theta) = \kappa_{\mathrm{ref}} u_0 \left(\frac{T_1}{T_{\mathrm{ref}}} \right)^{\alpha} \exp\left(\alpha \mu_{\mathrm{red}} \theta\right), \tag{11}$$

$$q_{\mathrm{res}}(\theta) = \mu_s T_1 \exp\left(\mu_{\mathrm{red}} \theta\right). \tag{12}$$

For $\alpha < 1$, the frictional resistance increases with cumulative wrap angle more rapidly than the vibration-induced loading. A vibration-mobilised region can therefore occur near the held end, where $q_{\mathrm{vib}} > q_{\mathrm{res}}$, while farther towards the loaded end $q_{\mathrm{vib}} < q_{\mathrm{res}}$. The ideal transition position is defined by

$$q_{\mathrm{vib}}(\theta^*) = q_{\mathrm{res}}(\theta^*). \tag{13}$$

This crossover represents a local mechanical threshold. The experimental transition may occur over a finite angular width, with θ^* identified as its centre.

2.4 Derivation of the Transition-Position Law

Substituting Eqs. 11 and 12 into the crossover condition, Eq. 13, gives

$$\kappa_{\mathrm{ref}} u_0 \left(\frac{T_1}{T_{\mathrm{ref}}} \right)^{\alpha} \exp\left(\alpha \mu_{\mathrm{red}} \theta^*\right) = \mu_s T_1 \exp\left(\mu_{\mathrm{red}} \theta^*\right). \tag{14}$$

Rearranging gives

$$\frac{\kappa_{\mathrm{ref}} u_0}{\mu_s T_{\mathrm{ref}}^{\alpha} T_1^{1-\alpha}} = \exp\left[(1-\alpha)\mu_{\mathrm{red}}\theta^*\right]. \tag{15}$$

Taking the natural logarithm,

$$\ln\left[\frac{\kappa_{\mathrm{ref}} u_0}{\mu_s T_{\mathrm{ref}}^{\alpha} T_1^{1-\alpha}}\right] = (1-\alpha)\mu_{\mathrm{red}}\theta^*. \tag{16}$$

Hence, for $\alpha < 1$,

$$\boxed{\theta^* = \frac{1}{(1-\alpha)\mu_{\mathrm{red}}} \ln\left[\frac{\kappa_{\mathrm{ref}} u_0}{\mu_s T_{\mathrm{ref}}^{\alpha} T_1^{1-\alpha}}\right]}. \tag{17}$$

The logarithmic dependence arises from the exponential spatial growth of the effective resistance profile together with the measured power-law dependence of tangential stiffness on tension.

2.5 Physical Regimes and Testable Predictions

The predicted transition lies within the wrapped contact only when $0 < \theta^* < \Theta$. Define the ratio of vibration-induced loading to local frictional resistance as

$$\mathcal{R}(\theta) = \frac{q_{\mathrm{vib}}(\theta)}{q_{\mathrm{res}}(\theta)} = \frac{\kappa_{\mathrm{ref}} u_0}{\mu_s T_1} \left(\frac{T_1}{T_{\mathrm{ref}}}\right)^{\alpha} \exp\left[-(1-\alpha)\mu_{\mathrm{red}}\theta\right]. \tag{18}$$

For $\alpha < 1$, $\mathcal{R}(\theta)$ decreases with increasing wrap angle. Three regimes therefore follow. If $\mathcal{R}(0) \leq 1$, no vibration-mobilised region is predicted. If $\mathcal{R}(0) > 1$ and $\mathcal{R}(\Theta) < 1$, the two force scales cross within the contact and $0 < \theta^* < \Theta$. If $\mathcal{R}(\Theta) \geq 1$, the vibration-induced loading exceeds the local resistance throughout the modelled wrap.

The corresponding threshold amplitudes are

$$u_{0,\mathrm{th}} = \frac{\mu_s T_1}{\kappa_{\mathrm{ref}}} \left(\frac{T_{\mathrm{ref}}}{T_1}\right)^{\alpha}, \tag{19}$$

$$u_{0,\mathrm{full}} = u_{0,\mathrm{th}} \exp\left[(1-\alpha)\mu_{\mathrm{red}}\Theta\right]. \tag{20}$$

Equation 19 is the same quantity introduced later in Eq. 45, written in ratio form. Within the partial-mobilisation regime, the transition law may be written as

$$\theta^* = \frac{1}{(1-\alpha)\mu_{\mathrm{red}}} \ln u_0 + C, \tag{21}$$

where C contains the quantities held fixed within an experimental series. Hence,

$$\frac{\partial\theta^*}{\partial \ln u_0} = \frac{1}{(1-\alpha)\mu_{\mathrm{red}}}, \qquad \frac{\partial\theta^*}{\partial \ln T_1} = -\frac{1}{\mu_{\mathrm{red}}}. \tag{22}$$

An increase in the amplitude of the vibration should shift the transition point towards the loaded end of the rope. An increase in the tension at the held end of the rope should shift the transition point towards the held end. The frequency of the vibration is not a parameter of the reduced model. Thus, for a given displacement amplitude u_0, the transition point should be independent of vibration frequency.

3.0 Experimental Methodology and Data Analysis

The methodology for performing the experiment separates the processes of model calibration and transition measurement. The quantities that enter into the theoretical prediction of the location of the transition between creeping and gripping regions are each measured prior to the vibration experiments, and high-speed imaging is performed to measure the cycle-accumulated relative displacement between the capstan and the rope along the wrapped rope.

The spatial profiles of creep rate along the wrapped rope are analysed using competing physical models of the responses of the rope to vibration. Only transitions that meet predefined model-selection, signal, and stability criteria are assigned a value of the transition angle, θ^*. Each of these measured transition locations is then compared to the theoretical prediction that results from the calibration of the model.

3.1 Experimental Strategy

The investigation consisted of two tests. The first test was to classify each of the measured spatial responses to determine whether they were consistent with a stable held-end creep-grip transition or with some alternative spatial response. For each of these spatial response classifications, a value of transition position, θ^*, was only reported if the model of a stable transition satisfied the predefined requirements based on the spatial response observations.

The second test employed the conditions that were classified as stable transitions to test the theoretical prediction of the position of the transition, θ^*. Each of the variables that described the experimental model - the cylinder radius r, the static friction coefficient μ_s, the reduced friction coefficient μ_{red}, and the tangential stiffness relation $\kappa_t(T)$ - was measured prior to the experiment, and without fitting to the measured positions of the transitions.

The experimental sequence was therefore

$$\begin{aligned}\text{calibration} \rightarrow \text{vibration experiment} \rightarrow \text{relative-displacement extraction} \\ \rightarrow \text{transition identification} \rightarrow \text{prediction test.}\end{aligned} \tag{23}$$

The principal observable was the cycle-accumulated relative displacement rate,

$$v_c(\theta) = \frac{d\Delta x_{\mathrm{rel}}(\theta, N)}{dN}, \tag{24}$$

where N is vibration-cycle number and $\Delta x_{\mathrm{rel}}(\theta, N)$ is the accumulated displacement of the rope relative to the adjacent capstan surface at cumulative angular position θ.

A gripping region was identified when $|v_c(\theta)|$ remained within the independently measured creep-rate floor σ_v, whereas a creeping region exhibited reproducible accumulated relative motion. Conditions for which no stable transition was identified were retained as experimental outcomes and were not assigned an artificial value of θ^*.

3.2 Experimental Apparatus and Optical Measurement

The experimental apparatus is shown in Fig. 3. A 3.0 mm braided polyester rope was wrapped through four complete turns around a 25.0 mm-diameter 6061 aluminium capstan. The cylinder radius was therefore $r = 12.5$ mm and the total cumulative wrap angle was $\Theta = 8\pi$. Adjacent turns were axially separated to prevent overlap or inter-turn contact.

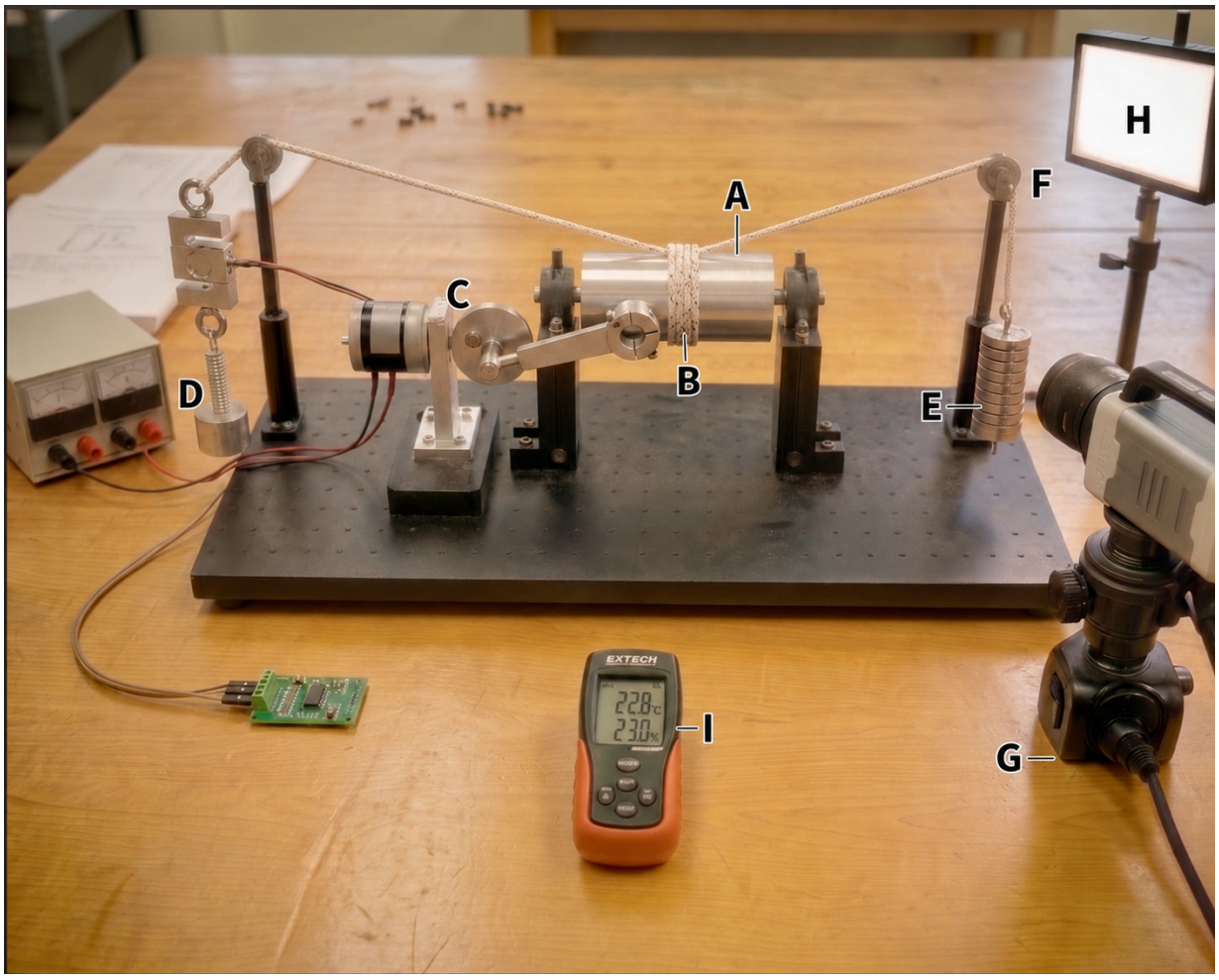


Figure 3: Experimental apparatus for the torsional-vibration measurements. **(A)** Aluminium capstan; **(B)** four-wrap braided polyester rope; **(C)** motor–cam–crank torsional drive; **(D)** held-end loading and force-measurement system defining T_1; **(E)** loaded-end dead-weight system defining T_2; **(F)** guide-pulley system; **(G)** high-speed camera; **(H)** continuous LED illumination; and **(I)** digital thermo-hygrometer.

The principal geometric, loading, drive, and acquisition specifications are summarized in Table 5 in Appendix B. Component labels in Fig. 3 are used only to identify the functional elements required to reproduce the measurement; commercially incidental mounting and support hardware are not individually specified. The rear-view mirrors used for optical registration of the direct and reflected camera views were positioned outside of the main photographic crop of the exposure and are shown schematically in Fig. 4. The direct and reflected camera views were calibrated independently using the measured capstan geometry and registered onto the cumulative coordinate defined in Eq. 1. Each of the rope ends was attached to loads via a guide pulley. The lightly loaded end of the rope was used to define T_1, while the loaded end defined T_2. The S-beam load cell attached to the system at T_1 was used to determine the mean held-end tension prior to and after each run. The force channel of the load cell was not used to determine the amplitude or phase of the vibration-frequency force.

The torsional excitation of the capstan was provided by a DC motor driving an eccentric cam connected through a rigid link to a crank fixed to the capstan shaft. The nominal motor setting was not used to determine the

vibration amplitude. Instead, an optical feature fixed to the capstan was tracked during each run to determine the angular trajectory of the capstan, $\phi(t)$.

The measured fundamental angular amplitude a then gave the surface-displacement amplitude

$$u_0 = ra. \tag{25}$$

A Photron high-speed camera was used to capture both the capstan motion and the rope motion. The high frame rate camera was operated at 1000 frames s^{-1} with an exposure time of approximately 0.20 ms using continuous LED illumination of the rope. The natural texture of the polyester rope was used to provide the optical means of tracking the rope motion, and an artificial speckle pattern was not required.

The same image sequence yielded the tangential displacement of the rope, $u_{\mathrm{rope}}(\theta, t)$, and the rigid-body tangential displacement of the capstan surface,

$$u_{\mathrm{cyl}}(t) = r\phi(t). \tag{26}$$

The local interfacial displacement was therefore obtained from

$$u_{\mathrm{rel}}(\theta, t) = u_{\mathrm{rope}}(\theta, t) - u_{\mathrm{cyl}}(t). \tag{27}$$

The four-wrap contact between the rope and the vibrating cylinder extends over both the portion directly visible to the camera and the rear portion of the cylinder. Consequently, the angular coordinate of the capstan is registered using both the direct camera images as well as the reflected images from the rear portion of the contact between the rope and cylinder. These two views have an overlapping angular range that is utilised to register both of those views onto the common coordinate of $0 \leq \theta \leq 8\pi$. This transformation mapping between the image plane and the angular coordinate is calibrated before testing the system to confirm the calibration and is retained as a fixed transformation for all other measurements.

Additionally, the temperature and relative humidity of the test environment are recorded during each experiment using a digital thermo-hygrometer.

3.3 Independent Calibration

All quantities entering the transition-position law were measured independently before the confirmatory values of θ^* were analysed. The model parameters were not fitted to the observed positions of the transition. The capstan geometry, friction coefficients, contact stiffness, vibration parameters, and optical measurement floors were all independently calibrated.

3.3.1 Geometry and Drive Calibration

The diameter of the capstan was measured with a digital caliper at a variety of positions around the capstan to determine the mean radius r of the capstan. The diameter of the rope was measured at several locations along the unloaded portion of the test spool. The direct and reflected camera views were calibrated using the measured capstan geometry and registered to the cumulative coordinate defined in Eq. 1. The mirrored reversal of the reflected camera images was corrected for prior to registration of the images, and the mismatch between the direct and reflected views within the angular overlap of the views was used as a measure of uncertainty in

the angular position. Furthermore, a feature was assigned to the capstan that was visible in the high-speed video sequences used to measure the angular position $\phi(t)$ of the capstan. The vibration frequency f and vibration amplitude a were determined directly from these measured high-speed video sequences.

The corresponding peak tangential displacement of the capstan surface was

$$u_0 = ra. \tag{28}$$

Measured optical values of a and f, rather than nominal motor settings, were used in all subsequent analysis.

3.3.2 Boundary Loading and Friction Calibration

The load-cell and pulley systems were calibrated using known masses. These calibrations allowed the actual values of T_1 and T_2 to be determined.

The static friction coefficient μ_s was measured separately using the same rope–aluminium materials in a calibration contact of wrap angle $\Theta_{\text{cal}} = \pi$. With no vibration applied, the tension at the loaded end was increased quasi-statically until relative motion between the rope and the drum was detected optically. The static friction coefficient was then calculated from

$$\mu_s = \frac{1}{\Theta_{\text{cal}}} \ln\left(\frac{T_{2,\text{crit}}}{T_{1,\text{crit}}}\right). \tag{29}$$

Measurements were repeated over the range of held-end tensions used in the confirmatory experiment.

The reduced friction coefficient μ_{red} was measured separately in the same calibration geometry during slow sustained relative motion. For each quasi-steady interval,

$$\mu_{\text{red}} = \frac{1}{\Theta_{\text{cal}}} \ln\left(\frac{T_{2,\text{creep}}}{T_{1,\text{creep}}}\right). \tag{30}$$

The calibration was repeated across the low-creep regime relevant to the vibration experiments, and the resulting value of μ_{red} was fixed before transition-position analysis.

3.3.3 Tangential Contact Stiffness

The effective preslip tangential stiffness was measured independently using a short rope-capstan contact. The reference calibration contact had angular extent

$$\Delta\theta = \frac{\pi}{6}. \tag{31}$$

At each prescribed baseline tension, small differential-force increments were applied while remaining below the onset of irreversible slip. The corresponding reversible displacement of the rope relative to the capstan was measured optically.

The finite-contact stiffness was obtained from the reversible portion of the force-displacement response,

$$K_t(T) = \frac{d(\Delta T)}{du_{\text{rel}}}, \tag{32}$$

and converted to stiffness per unit wrap angle as

$$\kappa_t(T) = \frac{K_t(T)}{\Delta\theta}. \tag{33}$$

The assumption in Eq. 33 is that the stiffness of the frictional contact is extensive over the short contact used for calibration. This was verified through calibrations with different wrap angles around the drum. The contact stiffness was found to be approximately linearly related to the wrap angle $\Delta\theta$, and the local stiffness per unit wrap angle was calculated.

Measurements at several baseline tensions spanning the experimental range were then represented by the power-law relation

$$\kappa_t(T) = \kappa_{\mathrm{ref}} \left(\frac{T}{T_{\mathrm{ref}}} \right)^{\alpha}, \qquad T_{\mathrm{ref}} = 1.00 \text{ N}. \tag{34}$$

Here, κ_{ref} is the fitted tangential stiffness per unit wrap angle at the fixed reference tension $T_{\mathrm{ref}} = 1.00$ N. The parameters κ_{ref} and α were determined entirely from the preslip calibration measurements and were not adjusted using the measured transition positions. The observed approximate proportionality $K_t \propto \Delta\theta$ supports the extensive-stiffness approximation used to express the vibration-induced force scale locally per unit wrap angle in Eq. 11. This approximation does not imply that neighbouring contact elements are dynamically uncoupled; rather, it provides an experimentally supported reduced-order conversion from the measured finite-contact preslip stiffness to the local stiffness scale used in the transition model.

3.3.4 Optical Measurement Floors

The optical measurement floor was determined using control recordings processed with the same tracking software as the vibration experiments. Both stationary recordings and oscillating conditions in which the rope remained gripping were used.

The apparent relative-displacement error was defined as

$$\epsilon_{\mathrm{rel}} = \epsilon_{\mathrm{rope}} - \epsilon_{\mathrm{cyl}}, \tag{35}$$

and its standard deviation defined the displacement floor,

$$\sigma_{\mathrm{opt}} = \mathrm{SD}\left(\epsilon_{\mathrm{rel}}\right). \tag{36}$$

Applying the same phase-matched displacement-versus-cycle analysis to the gripping controls gave the corresponding creep-rate floor,

$$\sigma_v = \mathrm{SD}\left(v_c^{\mathrm{gripped}}\right). \tag{37}$$

After completion of these calibrations, the values and uncertainties of r, μ_s, μ_{red}, κ_{ref}, α, σ_{opt}, and σ_v, together with the optical registration and load calibration, were fixed before confirmatory transition analysis.

3.4 Confirmatory Experimental Design

The confirmatory experiment was designed to test the predicted dependence of the transition position on vibration amplitude and held-end tension, as well as the predicted absence of explicit frequency dependence at fixed displacement amplitude. All model parameters used to define the experimental conditions were taken from the independent calibrations described in Section 3.3.

A reference condition was selected with the predicted transition near the first complete revolution of the contact,

$$\theta^*_{\mathrm{ref}} = \pi, \tag{38}$$

so that the transition lay well inside the measurable domain and away from either boundary. For a chosen reference held-end tension $T_{1,\mathrm{ref}}$, the corresponding reference displacement amplitude followed directly from Eq. 17,

$$u_{0,\mathrm{ref}} = \frac{\mu_s T_{\mathrm{ref}}^{\alpha} T_{1,\mathrm{ref}}^{1-\alpha}}{\kappa_{\mathrm{ref}}} \exp\left[(1-\alpha)\mu_{\mathrm{red}}\pi\right]. \tag{39}$$

The loaded-end tension T_2 was chosen so that the intended partial-transition regime was mechanically accessible while the remainder of the wrap remained below the vibration-off gross-slip limit. For a predicted transition at θ^*, this required

$$T_1 e^{\mu_{\mathrm{red}}\theta^*} < T_2 < T_1 \exp\left[\mu_{\mathrm{red}}\theta^* + \mu_s(\Theta - \theta^*)\right]. \tag{40}$$

Within the reduced model, T_2 does not determine the local crossover position directly. Instead, the loaded-end tension constrains whether a partial-mobilisation state compatible with the predicted θ^* can exist over the finite wrap. Thus, T_2 determines the admissible loading regime, whereas the transition-position law itself is parameterised by the held-end resistance scale T_1. This separation applies to the loading family examined here and should not be interpreted as a general statement that loaded-end tension is dynamically irrelevant. A fixed loaded-end to held-end tension ratio of

$$\boxed{\frac{T_2}{T_1} = 5.00} \tag{41}$$

was used throughout the amplitude and tension sweeps. Consequently, T_2 varied proportionally with T_1, from 3.55 N at $T_1 = 0.71$ N to 7.05 N at $T_1 = 1.41$ N. The same ratio was retained for the reference and frequency-test conditions.

3.4.1 Amplitude and Holding-Tension Sweeps

At fixed $T_{1,\mathrm{ref}}$, the surface-displacement amplitude was varied over five approximately logarithmically spaced levels,

$$\frac{u_0}{u_{0,\mathrm{ref}}} \in \{0.63,\ 0.80,\ 1.00,\ 1.26,\ 1.58\}. \tag{42}$$

This series tested the predicted linear dependence of θ^* on $\ln u_0$.

A second series varied the held-end tension while maintaining $u_0 = u_{0,\mathrm{ref}}$,

$$\frac{T_1}{T_{1,\mathrm{ref}}} \in \{0.71,\, 0.84,\, 1.00,\, 1.19,\, 1.41\}\,. \tag{43}$$

The loaded-end tension was adjusted proportionally so that the calibrated ratio T_2/T_1 remained unchanged. This series tested the predicted movement of the transition towards the held end as T_1 increased.

Because the two sweeps shared the reference condition, they contained nine unique primary operating conditions. Each condition was repeated six times using independently prepared rope-capstan contact regions, giving

$$9 \times 6 = 54 \tag{44}$$

primary measurements. A fresh four-wrap rope region was used for each run, and the order of the nine operating conditions was randomized within each repeat block.

3.4.2 Control and Frequency Tests

Two control conditions were included to test whether the optical and classification procedures generated an apparent transition in the absence of a physically mobilised region. The first used the reference loading with the vibration drive disabled. The second used a vibration amplitude below the predicted onset threshold.

Setting $\theta^* = 0$ in Eq. 17 gives the onset displacement amplitude

$$u_{0,\mathrm{th}} = \frac{\mu_s T_{\mathrm{ref}}^{\alpha} T_{1,\mathrm{ref}}^{1-\alpha}}{\kappa_{\mathrm{ref}}}, \tag{45}$$

and the sub-threshold control was performed at

$$u_{0,\mathrm{control}} = 0.50\, u_{0,\mathrm{th}}. \tag{46}$$

Six independent repeats were acquired for each control.

The final experiment tested whether the transition depended explicitly on excitation frequency. The reference loading and measured displacement amplitude were held fixed while the excitation frequency was varied over

$$f \in \{30,\, 40,\, 50,\, 60\}\ \mathrm{Hz}. \tag{47}$$

Because the 30 Hz reference condition was already included in the primary experiment, the three additional frequencies contributed 18 additional measurements.

The complete experimental campaign therefore contained

$$54 + 12 + 18 = 84 \tag{48}$$

physical runs: 54 primary amplitude–tension measurements, 12 control measurements, and 18 additional frequency-test measurements. Each physical run followed the same chronological protocol: the apparatus was allowed to settle for 60 s, followed by a 5 s pre-run measurement of T_1, a 1 s stationary optical baseline, 2 s of drive spin-up, at least 5 s of high-speed recording, and a 5 s post-run measurement of T_1. The rope was then

advanced to a fresh four-wrap contact region before the next run. The corresponding run-retention checks and acquisition details are summarized in Appendix B.

3.5 Image Processing and Creep-Rate Extraction

High-speed recordings were processed to recover the motion of the capstan and the rope on a common cumulative angular coordinate. The objective of the analysis was to remove the reversible oscillatory motion of the cylinder and isolate the cycle-accumulated displacement of the rope relative to the adjacent capstan surface.

A rigid feature fixed to the capstan was tracked in each frame to obtain the angular trajectory $\phi(t)$. Successive positive extrema of $\phi(t)$ were used as the phase reference for cycle matching. Rope positions were then sampled at the same phase on selected cycles,

$$N = 0, 5, 10, 15, \ldots, 150, \tag{49}$$

so that the reversible within-cycle displacement was not interpreted as accumulated creep.

Natural braid texture was used to track the rope at discrete cumulative positions θ_i along the four-wrap contact. For each sampled frame, the tangential displacement of the rope and the corresponding rigid-body displacement of the capstan surface were determined from the calibrated image coordinates. Their difference defined the local relative displacement,

$$u_{\mathrm{rel}}(\theta_i, N) = u_{\mathrm{rope}}(\theta_i, N) - u_{\mathrm{cyl}}(N). \tag{50}$$

Relative displacement was referenced to the first accepted phase-matched sample,

$$\Delta x_{\mathrm{rel}}(\theta_i, N) = u_{\mathrm{rel}}(\theta_i, N) - u_{\mathrm{rel}}(\theta_i, 0). \tag{51}$$

At each spatial position, the accumulated displacement was fitted as a function of cycle number,

$$\Delta x_{\mathrm{rel}}(\theta_i, N) = b_i + v_c(\theta_i)N + \epsilon_i, \tag{52}$$

where b_i is an intercept and ϵ_i denotes the residual measurement error. The fitted slope,

$$v_c(\theta_i) = \frac{d\Delta x_{\mathrm{rel}}(\theta_i, N)}{dN}, \tag{53}$$

defined the cycle-accumulated creep rate at that position.

Repeating this procedure over the full contact produced the spatial creep-rate profile

$$v_c(\theta), \qquad 0 \le \theta \le 8\pi. \tag{54}$$

The sign of the creep rate, v_c, was retained during the fitting and comparison of model profiles. The magnitude of the computed creep rate, $|v_c(\theta)|$, was plotted to provide a visualization of the spatial profile. Any regions in which the magnitude remained within the range of the independently calibrated creep-rate floor, σ_v, were considered to be gripping within the resolution of the measurement system.

The complete extraction workflow is summarized in Fig. 4.

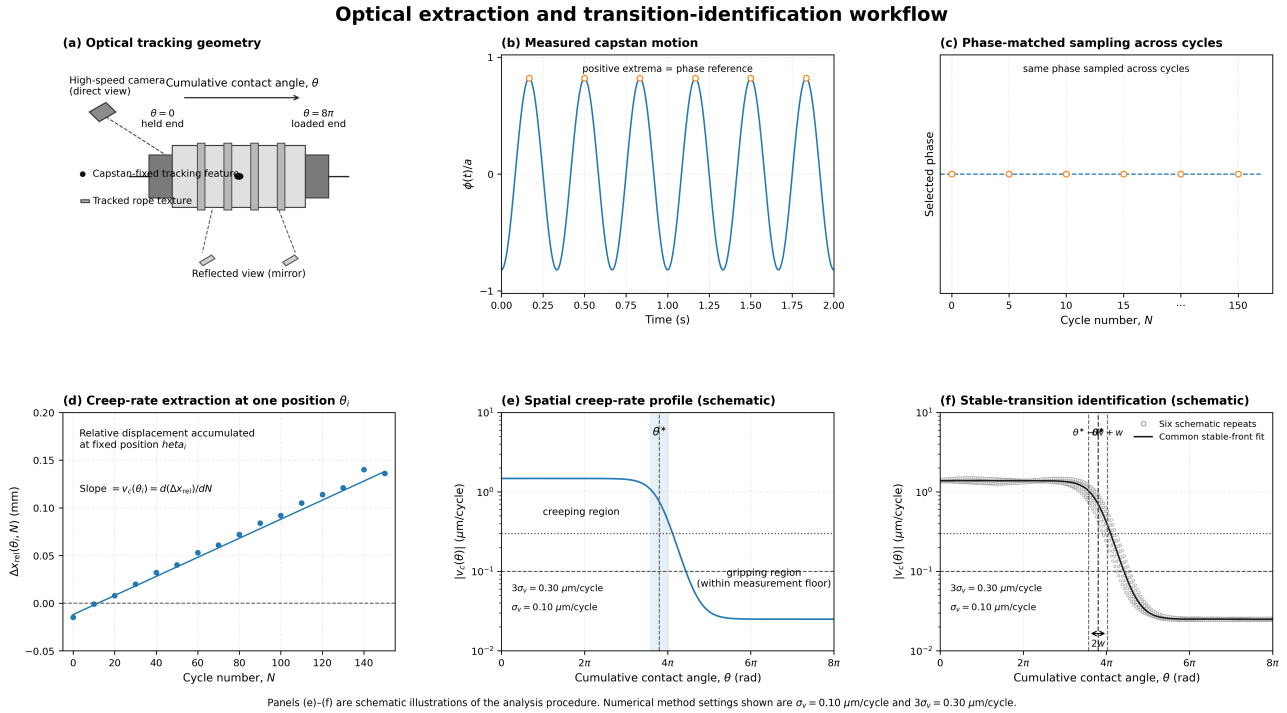


Figure 4: Optical extraction and transition-identification workflow. (a) Optical tracking geometry for the four-wrap contact, with cumulative coordinate θ measured from the held end to the loaded end. (b) Capstan angular motion $\phi(t)$ obtained from a rigid capstan-fixed feature; successive positive extrema define the phase reference. (c) Phase-matched sampling at equivalent mechanical states on selected cycles. (d) At a fixed position θ_i, linear regression of accumulated relative displacement against cycle number gives the local creep rate $v_c(\theta_i)$. (e) Schematic processing illustration showing the assembled spatial creep-rate magnitude $|v_c(\theta)|$ and the independently determined measurement floor σ_v. (f) Schematic illustration of six repeated profiles and the stable-transition fitting procedure. The fitted centre is θ^*; the dashed outer lines indicate $\theta^* - w$ and $\theta^* + w$, so the displayed interval has width $2w$.

Panels (d)–(f) in Fig. 4 present representative examples of each of the steps in the extraction process, included only to provide a representation of the extraction and fitting procedure; the actual experimental results are presented separately in Section 4.0. The direct and reflected image regions were mapped onto the same cumulative coordinate prior to the construction of the profiles. Those features within the calibrated region of overlap between the direct and reflected views were utilized to confirm the registration of the two views, and any features that could not be localized within the image were excluded from the fit. No interpolation methods were used in the construction of the profiles.

Each physical condition contributed to the formation of one profile per experimental repeat. Each profile was separately retained, rather than averaged across repeats, for the model comparison procedure.

3.6 Transition Identification

A creep-grip transition was not assumed from the appearance of an individual spatial profile. For each experimental condition, the six measured creep-rate profiles were fitted jointly to five competing spatial-response models, representing distinct physical behaviours. These models are summarized in Table 1 and illustrated schematically in Fig. 5.

Model A contained one constant offset for each repeat and therefore described conditions with no resolved spatial dependence. Model B represented a smooth distributed response using

Table 1: Candidate spatial-response models used for transition identification.

Model	Physical interpretation
A	No resolved spatial structure
B	Smooth distributed creep without a localized transition
C_{stable}	Localized transition with a common centre across repeats
$C_{\mathrm{migrating}}$	Localized transition with repeat-dependent centre
D	Multiple localized slip regions

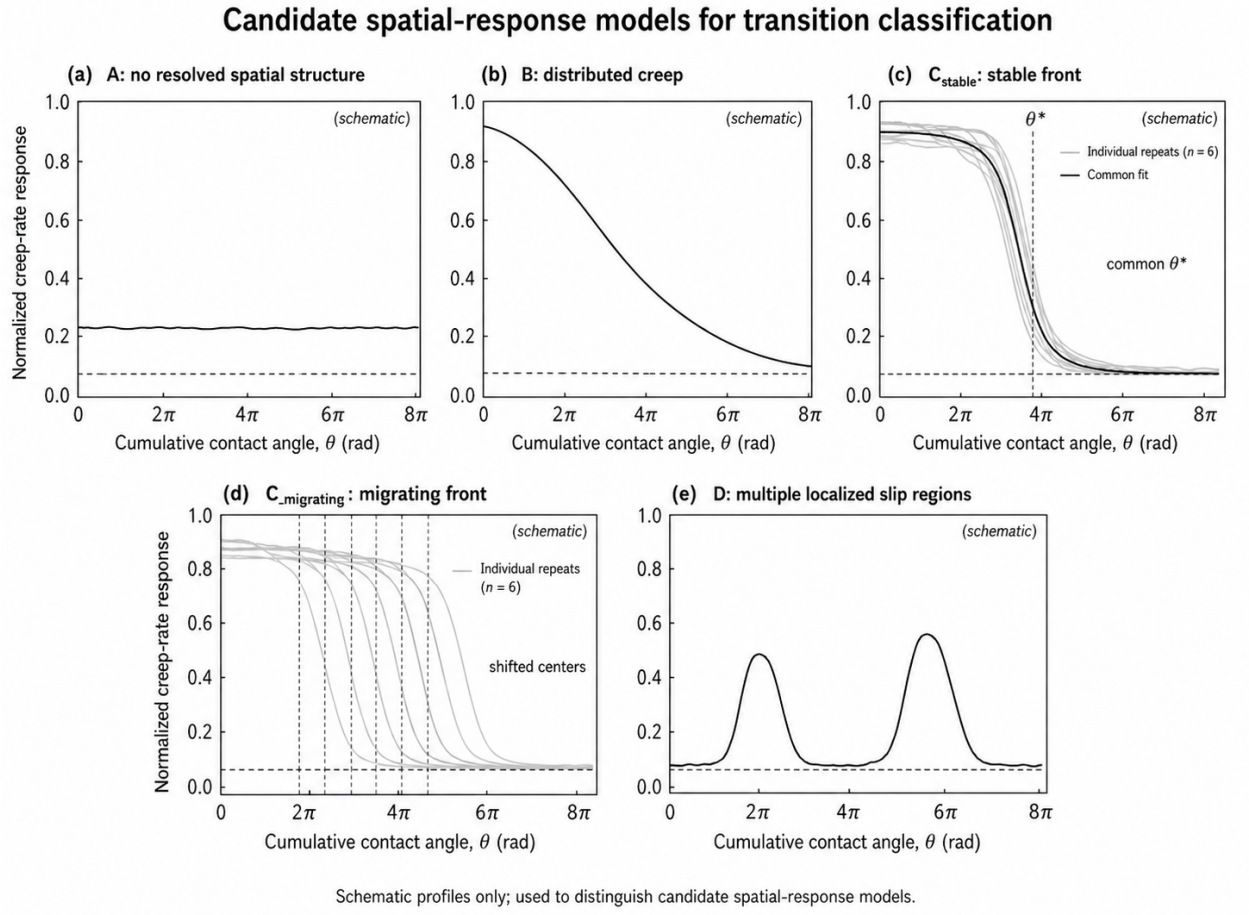


Figure 5: Schematic spatial-response models used for transition classification. (a) Model A represents a spatially uniform response with no resolved angular structure, with the offset allowed to vary between repeats. (b) Model B represents smooth distributed creep varying monotonically across the contact without a localized transition centre. (c) C_{stable} represents a localized front with a centre and width shared across all six repeats, while repeat amplitudes and offsets may differ. (d) $C_{\mathrm{migrating}}$ allows the transition centre and width to vary between repeats. (e) Model D represents multiple localized slip regions using two positive spatial patches. The profiles are schematic and illustrate the mathematical alternatives used for model comparison rather than experimental results.

$$v_{c,i}(\theta) = b_i + A_i \left[1 - \left(\frac{\theta}{\Theta}\right)^p\right]. \tag{55}$$

where b_i and A_i are repeat-specific offset and amplitude terms and p controls the distributed spatial decay. Unlike a front model, Model B contains no transition centre.

The stable-transition model was

$$v_{c,i}(\theta) = b_i + A_i \left[\frac{1}{2}\left(1 - \tanh\left(\frac{\theta - \theta^*}{w}\right)\right)\right], \tag{56}$$

where θ^* and w are shared across all six repeats, while b_i and A_i remain repeat-specific. The migrating-front model used the same spatial form but allowed the transition centre and width to vary independently between repeats.

Model D provided an explicit non-front alternative for profiles containing multiple localized regions. Each repeat was represented by two positive Gaussian-like patches,

$$v_{c,i}(\theta) = b_i + A_{1,i} \exp\left[-\left(\frac{\theta - c_{1,i}}{s_{1,i}}\right)^2\right] + A_{2,i} \exp\left[-\left(\frac{\theta - c_{2,i}}{s_{2,i}}\right)^2\right], \tag{57}$$

with constrained patch widths and separation. This prevented two localized slip regions from being incorrectly interpreted as a single monotonic front.

3.6.1 Model Selection and Front Criteria

Primary transition classification used six experimental repeats represented on a common grid of 120 predefined cumulative angular bins over $0 \leq \theta \leq 8\pi$, corresponding to a nominal angular spacing of approximately 12°. Missing optical measurements remained missing and were not replaced by zero or interpolated for inference. At least three valid repeat measurements were required to contribute to an accepted angular bin. All candidate models were fitted to the same set of valid measurements.

Because errors at neighbouring angular positions were not assumed to be independent, first-order autoregressive prewhitening was applied to the residuals before model comparison. Model support was quantified using the Bayesian Information Criterion,

$$\mathrm{BIC} = n \ln\left(\frac{\mathrm{RSS}}{n}\right) + k \ln n, \tag{58}$$

where n is the number of valid observations, k is the number of fitted parameters, and RSS is the residual sum of squares after prewhitening.

A stable front was required to exceed the best non-front explanation by

$$\Delta\mathrm{BIC}_{\mathrm{stable}} = \min\left(\mathrm{BIC}_A, \mathrm{BIC}_B, \mathrm{BIC}_D\right) - \mathrm{BIC}_{C_{\mathrm{stable}}} \geq 10. \tag{59}$$

Model preference alone was not sufficient to report a transition. The fitted front amplitude was also required to exceed three times the independently measured creep-rate floor, and the fitted centre had to remain at least

$$0.35 \text{ rad} \tag{60}$$

from either boundary of the measured angular domain.

The migrating-front model was evaluated separately. A response was classified as migrating rather than stable when

$$\mathrm{BIC}_{C_{\mathrm{stable}}} - \mathrm{BIC}_{C_{\mathrm{migrating}}} \geq 10, \tag{61}$$

so that the additional repeat-dependent transition parameters were accepted only when they produced decisive improvement in model evidence.

Multiple localized structure was also checked explicitly. Profiles containing two sufficiently resolved spatial peaks, or residual non-monotonic structure after fitting a single front, were retained as irregular or multi-patch responses rather than forced into C_{stable}.

3.6.2 Stability of the Transition Estimate

A provisional stable-front classification was subjected to additional repeat-consistency, resampling, and deletion tests before a value of θ^* was reported.

For the six physical repeats, the dispersion of the independently fitted repeat centres was defined as

$$s_c = \sqrt{\frac{1}{J-1}\sum_{j=1}^{J}(c_j - \overline{c})^2}, \qquad J = 6. \tag{62}$$

A stable transition required

$$s_c \leq 0.105 \text{ rad} \approx 6^\circ. \tag{63}$$

Repeat-level bootstrap datasets were generated by resampling the six complete physical repeats with replacement. Spatial-block bootstrap datasets were generated separately by resampling contiguous blocks of neighbouring angular bins, thereby preserving the local spatial correlation present in the measured profiles. The complete model-comparison and classification procedure was reapplied to every resampled dataset.

For a transition to be accepted, the bootstrap standard deviation of the fitted centre was required to satisfy

$$\mathrm{SD}_{\mathrm{boot}}(\theta^*) \leq 0.14 \text{ rad}, \tag{64}$$

and the relative bootstrap standard deviation of the fitted width was required to satisfy

$$\frac{\mathrm{SD}_{\mathrm{boot}}(w)}{\overline{w}_{\mathrm{boot}}} \leq 0.35. \tag{65}$$

The complete analysis was additionally repeated after removing each physical repeat individually and after removing each valid angular bin individually. The maximum allowed changes in the fitted centre were

$$\max |\Delta\theta^*_{\text{leave-one-bin}}| \leq 0.14 \text{ rad}, \tag{66}$$

and

$$\max |\Delta\theta^*_{\text{leave-one-repeat}}| \leq 0.18 \text{ rad.} \tag{67}$$

Finally, the stable-front classification was required to change in no more than 5% of the deletion analyses.

A value of θ^* was therefore reported only when the fitted response satisfied the model-evidence, signal, edge-clearance, repeat-consistency, migration, multi-patch, bootstrap, and deletion-stability requirements. Conditions that failed any of these safeguards remained recorded experimental outcomes but were not assigned an artificial transition position.

3.7 Validation of the Predictive Law and Uncertainty Analysis

Only those experimental conditions classified as stable transitions in Section 3.6 were assigned a measured value of the transition position, θ^*. Each of these measured values was compared with the value of the transition position that was predicted for those same conditions (as determined in Section 2.0),

$$\theta^*_{\text{pred}} = \frac{1}{(1-\alpha)\mu_{\text{red}}} \ln\left[\frac{\kappa_{\text{ref}} u_0}{\mu_s T_{\text{ref}}^{\alpha} T_1^{1-\alpha}}\right]. \tag{68}$$

Each of the quantities appearing on the right-hand side of this equation was obtained from calibrations independent of the determination of the transition position (as described in Section 3.3); no measured value of the transition position was used to determine the value of either κ_{ref}, α, μ_s, or μ_{red}.

3.7.1 Amplitude and Tension Dependence

The amplitude series was analysed by regressing the measured transition positions against the value of $\ln u_0$ for each experiment. The result of this regression can be written as

$$\theta^* = \beta_a \ln u_0 + C_a, \tag{69}$$

where the experimentally determined value of the slope, β_a, was compared to the value of the slope predicted for this relationship in Section 2.5. The regression that was performed was a weighted least-squares regression, with weights proportional to the inverse variance of the measured position of the transition.

$$w_i = \frac{1}{\sigma^2_{\theta^*,i}}. \tag{70}$$

The holding-tension series was analysed independently using

$$\theta^* = \beta_T \ln T_1 + C_T, \tag{71}$$

and the fitted slope β_T was compared with the corresponding calibration-fixed theoretical prediction. Thus, these two regressions allowed the investigation of the predicted logarithmic dependence of the transition position without introducing any additional parameters into the physical law associated with the process.

3.7.2 Absolute Prediction Test

The predictive law was also tested in absolute rather than only differential form. For each stable-front condition, the independently measured values of u_0, T_1, μ_s, μ_{red}, κ_{ref}, and α were inserted into Eq. 68 to obtain θ^*_{pred} before comparison with the optically measured θ^*_{meas}.

Agreement was assessed using the residual

$$\delta\theta^* = \theta^*_{\mathrm{meas}} - \theta^*_{\mathrm{pred}}, \tag{72}$$

together with the absolute and root-mean-square prediction errors across the stable-front conditions. The measured and predicted positions were also compared using

$$\theta^*_{\mathrm{meas}} = \beta_0 + \beta_1 \theta^*_{\mathrm{pred}}, \tag{73}$$

for which agreement with the calibration-fixed law corresponds to $\beta_0 = 0$ and $\beta_1 = 1$ within experimental uncertainty. This comparison tested the absolute predictive capability of the model rather than only its ability to reproduce a trend.

3.7.3 Frequency Falsification Test

For these experiments, the frequency responses of the system were investigated to determine if the excitation frequency had any effect upon the transition law for a fixed value of the displacement amplitude. For each test run, each of the parameters u_0, T_1, and T_2/T_1 was held at its reference value, and the excitation frequency was varied.

A possible residual frequency dependence was quantified using

$$\theta^* = \beta_f \ln f + C_f. \tag{74}$$

According to the model that incorporated the displacement control parameter, the slope of the response to frequency variations was expected to be zero ($\beta_f = 0$). Thus, testing of the model through these experiments functioned as a falsification experiment; the detection of a non-zero slope for the response of θ^* to f would indicate that another rate-dependent process must contribute to the system.

3.7.4 Uncertainty Propagation

The uncertainty in the position of the transition was obtained through the repeat- and spatial-resampling procedures described in Section 3.6.2. For each experiment, the distribution of values of the transition centre accepted from each resampled fit to the experimental data provided a confidence interval for the value of θ^*_{meas}. The uncertainty of the predicted transition position was propagated independently from the measured transition-position uncertainty using the measured model inputs. Thus, the experimentally determined uncertainties in

$$r, \quad a, \quad T_1, \quad \mu_s, \quad \mu_{\mathrm{red}}, \quad \kappa_{\mathrm{ref}}, \quad \alpha \tag{75}$$

were sampled from their calibration distributions and substituted into Eq. 68. The resulting distribution of θ^*_{pred}

provided the model-prediction uncertainty while preserving the nonlinear dependence of the transition law on its measured parameters. Uncertainty in each of the experimentally fitted slopes of the amplitude, tension, and frequency series was determined by recalculating the slopes from bootstrap samples of the measured transition positions. The uncertainty in the corresponding theoretical slopes was propagated separately from the calibrated parameter distributions of μ_{red} and α, as applicable. The classification of each condition in the experiment and the theoretical determination of whether each lay within the predicted regime of partial mobilisation were separate analyses. Disagreement between the theoretical law and the experimental determination of the transition between creep and grip was not, therefore, a criterion for exclusion from the experiment. Conditions that produced no accepted stable transition remained part of the reported experimental campaign.

4.0 Results

4.1 Independent Calibration Results

All quantities entering the transition-position law were calibrated independently before analysis of θ^* and are summarized in Table 2.

Table 2: Independent calibration parameters used in the transition-position prediction. Uncertainties are 95% confidence intervals; $T_{\mathrm{ref}} = 1.00$ N.

Parameter	Physical quantity	Calibrated value	Method
r	Capstan radius	12.50 ± 0.03 mm	Geometric calibration
μ_s	Static friction coefficient	0.30 ± 0.02	Quasi-static breakaway
μ_{red}	Reduced friction coefficient	0.18 ± 0.02	Slow-creep calibration
κ_{ref}	Reference tangential stiffness	1.60 ± 0.12 kN m^{-1} rad^{-1}	Preslip calibration
α	Tension–stiffness exponent	0.45 ± 0.05	Power-law stiffness fit
σ_{opt}	Optical displacement floor	0.015 ± 0.003 mm	Stationary/gripping control
σ_v	Creep-rate floor	0.10 ± 0.02 μm cycle^{-1}	Phase-matched gripping control

The static coefficient of friction ($\mu_s = 0.30 \pm 0.02$) was greater than the coefficient measured under slow creep ($\mu_{\mathrm{red}} = 0.18 \pm 0.02$; Fig. 6(a)). Furthermore, the exponent α ($\alpha = 0.45 \pm 0.05$), which described the dependence of tangential contact stiffness on tension, was within the required range $0 < \alpha < 1$. Tests performed at different short-contact wrap angles indicated that K_t was approximately linearly related to the contact angle $\Delta\theta$, supporting the use of the ratio $\kappa_t = K_t/\Delta\theta$ for describing the local stiffness per unit wrap angle. Both the stationary and gripping controls gave an optical displacement floor of $\sigma_{\mathrm{opt}} = 0.015 \pm 0.003$ mm and a creep-rate floor of $\sigma_v = 0.10 \pm 0.02$ μm cycle^{-1}, respectively. The creep-rate floor defined the lower measurement limit for creep within the experiment, while $3\sigma_v$ provided the signal criterion used for classifying transitions (Fig. 6(c)). These calibrated quantities were held constant during the transition analysis and were not adjusted according to the measured values of the transition angle θ^*.

4.2 Observation and Classification of the Creep-Grip Transition

A representative above-threshold condition produced a localized transition from held-end creep to loaded-end gripping. Across all six repeats, the creep-rate magnitude was largest near the held end and decreased with cumulative contact angle. The held-side response remained above the independently calibrated signal threshold, $3\sigma_v = 0.30$ μm cycle^{-1}, whereas the loaded-side response approached the gripping floor, $\sigma_v = 0.10$ μm cycle^{-1}. The calibrated measurement and signal thresholds are shown in Fig. 6(c).

Joint fitting of the six repeats with the stable-front model gave

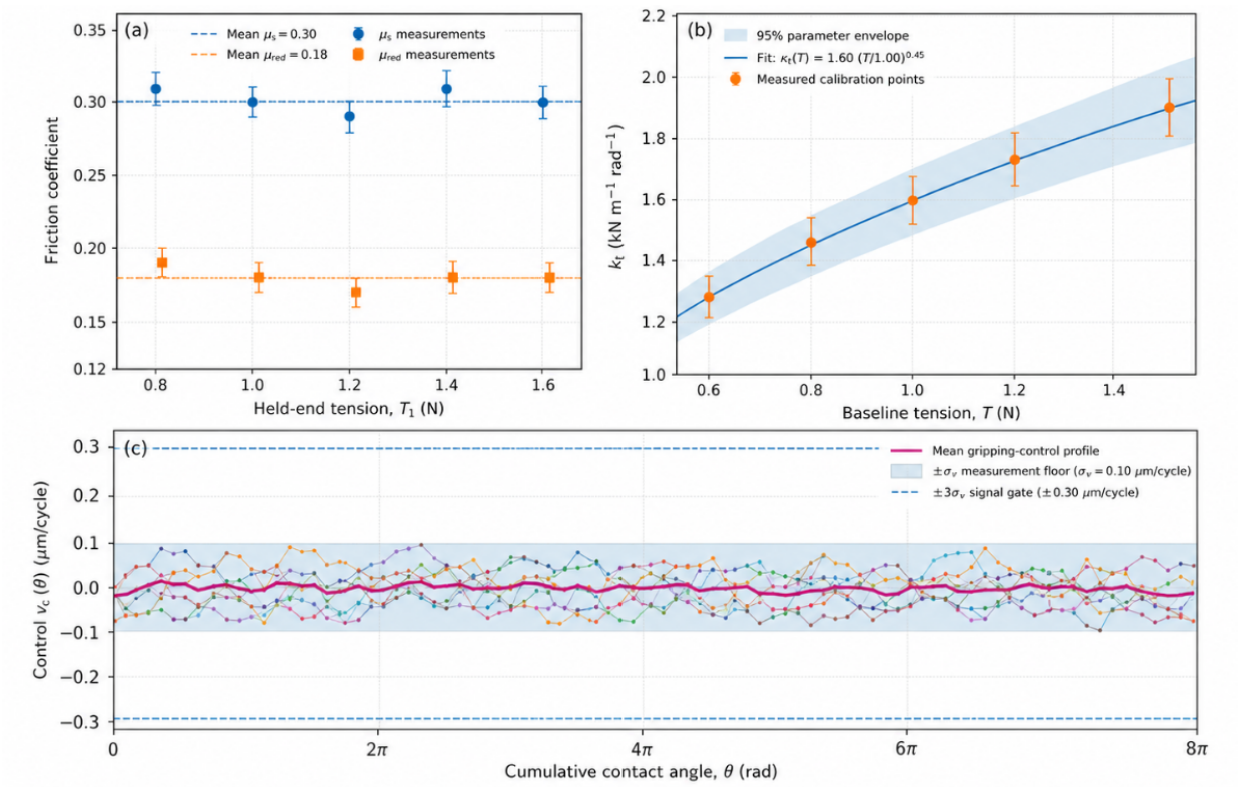


Figure 6: Independent calibration results. (a) Independently measured static and reduced slow-creep friction coefficients across the calibration tension range; dashed lines indicate their mean calibrated values. (b) Preslip tangential stiffness as a function of baseline tension and the fitted power law $\kappa_t(T) = \kappa_{\mathrm{ref}}(T/T_{\mathrm{ref}})^{\alpha}$; shading denotes the 95% propagated fit uncertainty. (c) Individual gripping-control creep-rate profiles and their mean, used to determine the calibrated $\pm\sigma_v$ measurement floor and corresponding $\pm 3\sigma_v$ signal criterion.

$$\theta^*_{\mathrm{meas}} = 3.27 \pm 0.17 \text{ rad}, \tag{76}$$

close to the independently targeted reference position π. The fitted transition had a finite spatial width rather than an abrupt discontinuity.

The stable-front model was decisively preferred over the non-front alternatives, with

$$\Delta\mathrm{BIC}_A = 31.8, \qquad \Delta\mathrm{BIC}_B = 19.6, \qquad \Delta\mathrm{BIC}_D = 16.9, \tag{77}$$

all exceeding the predefined threshold $\Delta\mathrm{BIC} = 10$. Allowing repeat-dependent front positions gave only $\Delta\mathrm{BIC}_{C_{\mathrm{migrating}}} = 5.2 < 10$ relative to C_{stable}, so the additional migrating-front parameters were not supported. The fit also satisfied the repeat-consistency, multi-patch, bootstrap, and deletion-stability criteria defined in Section 3.6.

Neither control condition produced an accepted transition. With vibration disabled ($u_0 = 0$), the response remained within the gripping floor throughout the contact. The sub-threshold control, $u_0 = 0.5u_{0,\mathrm{th}}$, likewise produced no accepted front. Both outcomes are reported in the complete condition-level summary in Table 4.

Torsional vibration therefore produced a reproducible localized creep–grip transition, whereas neither control condition produced an accepted front. The following subsection tests whether the measured transition position follows the predicted amplitude dependence.

4.3 Amplitude Dependence of the Transition Position

The five amplitude levels defined in Section 3.4.1 were

$$\frac{u_0}{u_{0,\mathrm{ref}}} = \{0.63,\ 0.80,\ 1.00,\ 1.26,\ 1.58\}\,, \qquad u_{0,\mathrm{ref}} = 0.2559 \text{ mm}. \tag{78}$$

The lowest-amplitude condition, $u_0 = 0.63u_{0,\mathrm{ref}}$, lay beneath the independently predicted onset of the transition and failed to produce an accepted stable front (Fig. 7). For the remaining four conditions, however, stable transitions were identified at

$$\theta^*_{\mathrm{meas}} = 1.03 \pm 0.17,\ 3.27 \pm 0.17,\ 5.29 \pm 0.18,\ 7.60 \pm 0.22 \text{ rad} \tag{79}$$

for $u_0/u_{0,\mathrm{ref}} = 0.80,\ 1.00,\ 1.26,\ 1.58$, respectively. Thus, the transition moved monotonically toward the loaded end of the rope with increasing displacement amplitude. Because the reference displacement $u_{0,\mathrm{ref}}$ was defined independently to give a transition position of $\theta^* = \pi$, the calibration-fixed prediction for the position of the transition was

$$\theta^*_{\mathrm{pred}} = \pi + \beta_{a,\mathrm{theory}} \ln\left(\frac{u_0}{u_{0,\mathrm{ref}}}\right), \qquad \beta_{a,\mathrm{theory}} = \frac{1}{(1-\alpha)\mu_{\mathrm{red}}}. \tag{80}$$

Using $\alpha = 0.45 \pm 0.05$ and $\mu_{\mathrm{red}} = 0.18 \pm 0.02$ gave

$$\boxed{\beta_{a,\mathrm{theory}} = 10.10 \pm 1.45 \text{ rad}} \tag{81}$$

at the 95% confidence level. Weighted least-squares regression of the four accepted fronts gave

$$\boxed{\beta_{a,\mathrm{meas}} = 9.54 \pm 0.74 \text{ rad}} \tag{82}$$

at the 95% confidence level. Thus, the measured and predicted slopes of the transition position versus displacement amplitude agreed within their propagated uncertainties.

The amplitude series therefore supports the prediction that the position of the transition is a logarithmic function of the imposed displacement amplitude.

4.4 Holding-Tension Dependence

The five held-end tension levels defined in Section 3.4.1 were

$$\frac{T_1}{T_{1,\mathrm{ref}}} = \{0.71,\ 0.84,\ 1.00,\ 1.19,\ 1.41\}\,, \tag{83}$$

with $u_0 = u_{0,\mathrm{ref}}$ and constant T_2/T_1.

All five conditions produced stable creep-grip transitions at

$$\theta^*_{\mathrm{meas}} = 5.10 \pm 0.15,\ 4.05 \pm 0.13,\ 3.27 \pm 0.17,\ 2.12 \pm 0.12,\ 1.28 \pm 0.14 \text{ rad}, \tag{84}$$

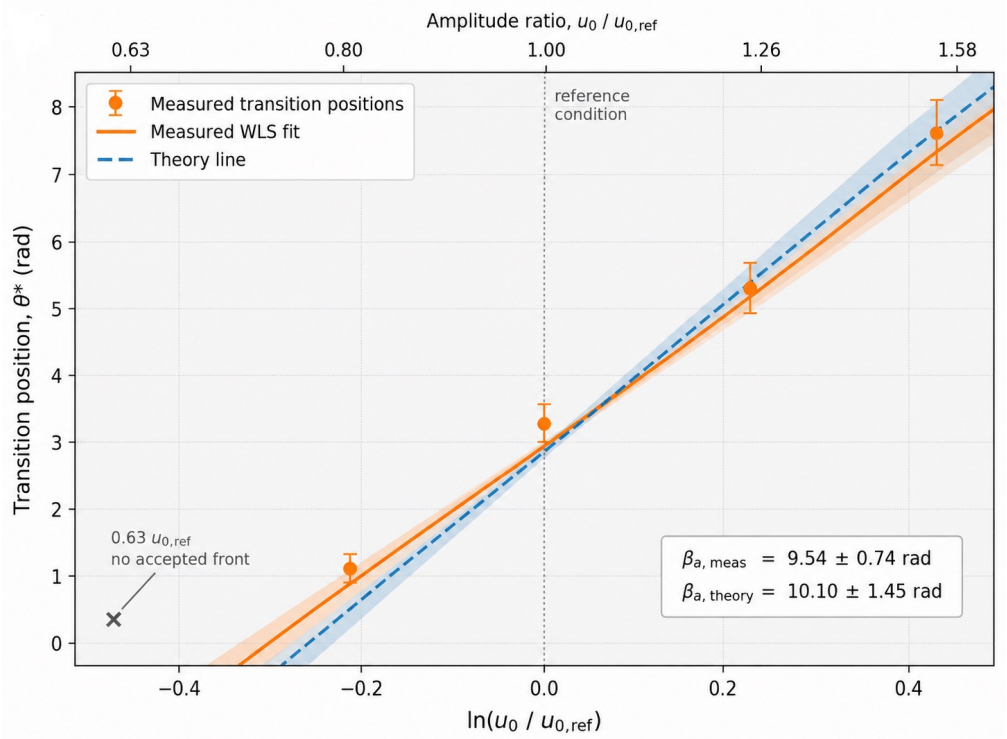


Figure 7: Amplitude dependence of the creep–grip transition position. Accepted stable fronts are plotted against $\ln(u_0/u_{0,\mathrm{ref}})$. The solid line is the weighted experimental fit, while the dashed line is the independently calibrated prediction anchored at $\theta^* = \pi$ for $u_0 = u_{0,\mathrm{ref}}$. Shaded regions denote the corresponding 95% uncertainty bands. The $0.63u_{0,\mathrm{ref}}$ condition lay below onset and produced no accepted transition; the cross marks this tested condition schematically and does not represent a measured value of θ^*.

for $T_1/T_{1,\mathrm{ref}} = 0.71,\ 0.84,\ 1.00,\ 1.19,\ 1.41$, respectively. Increasing T_1 therefore moved the transition monotonically toward the held end (Fig. 8).

For fixed displacement amplitude, the calibration-fixed prediction is

$$\theta^*_{\mathrm{pred}} = \pi + \beta_{T,\mathrm{theory}} \ln\left(\frac{T_1}{T_{1,\mathrm{ref}}}\right), \qquad \beta_{T,\mathrm{theory}} = -\frac{1}{\mu_{\mathrm{red}}}. \tag{85}$$

Using $\mu_{\mathrm{red}} = 0.18 \pm 0.02$ gave

$$\boxed{\beta_{T,\mathrm{theory}} = -5.56 \pm 0.62 \text{ rad}} \tag{86}$$

at the 95% confidence level. Weighted least-squares regression of the five measured transition positions gave

$$\boxed{\beta_{T,\mathrm{meas}} = -5.56 \pm 0.50 \text{ rad}} \tag{87}$$

at the 95% confidence level. The measured and predicted slopes therefore agreed within their propagated uncertainties.

The tension series therefore supports the predicted logarithmic dependence of the transition position on held-end tension.

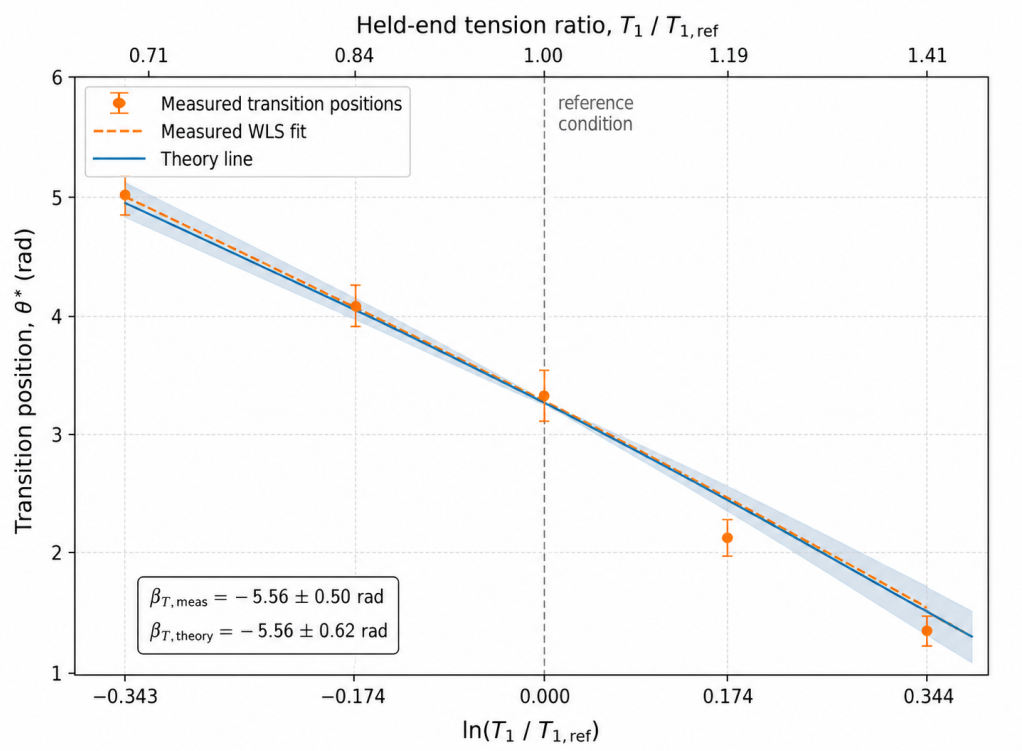


Figure 8: Held-end tension dependence of the creep–grip transition position. Accepted stable fronts are plotted against $\ln(T_1/T_{1,\mathrm{ref}})$. The dashed line is the weighted experimental fit, while the solid line is the independently calibrated prediction anchored at $\theta^* = \pi$ for $T_1 = T_{1,\mathrm{ref}}$. Shading denotes the propagated 95% theoretical uncertainty.

4.5 Absolute Prediction of the Transition Position

A further test is whether the independently calibrated law predicts the absolute position of the transition, and not only its trend, without fitting the model to the measured values of θ^*. For each of the accepted stable-front conditions,

$$\theta^*_{\mathrm{pred}} = \frac{1}{(1-\alpha)\mu_{\mathrm{red}}} \ln\left[\frac{\kappa_{\mathrm{ref}} u_0}{\mu_s T_{\mathrm{ref}}^{\alpha} T_1^{1-\alpha}}\right] \tag{88}$$

was evaluated using only independently calibrated quantities of the system. The four accepted conditions of the amplitude series and the four non-reference conditions of the tension series gave eight unique stable-front operating points.

Measured and predicted positions were compared using

$$\theta^*_{\mathrm{meas}} = \beta_0 + \beta_1 \theta^*_{\mathrm{pred}}, \tag{89}$$

for which exact agreement would be indicated by $\beta_0 = 0$ and $\beta_1 = 1$. For descriptive comparison with the identity relation, a weighted regression was performed using the measured uncertainties in each value of θ^*:

$$\beta_0 = 0.10 \pm 0.13 \text{ rad}, \qquad \beta_1 = 0.968 \pm 0.035, \tag{90}$$

at the 95% confidence level.

Across the eight conditions (Fig. 9),

$$\text{RMSE} = 0.117 \text{ rad}, \qquad \text{median}\,|\delta\theta^*| = 0.094 \text{ rad}, \tag{91}$$

where $\delta\theta^* = \theta^*_{\text{meas}} - \theta^*_{\text{pred}}$. Because the predicted positions also contain uncertainty from shared calibration parameters, this regression is used only as a descriptive identity check rather than as the primary statistical test of model agreement. The absolute prediction error is therefore assessed principally from the condition-wise residuals and the error metrics reported below, while the predicted uncertainties are retained separately.

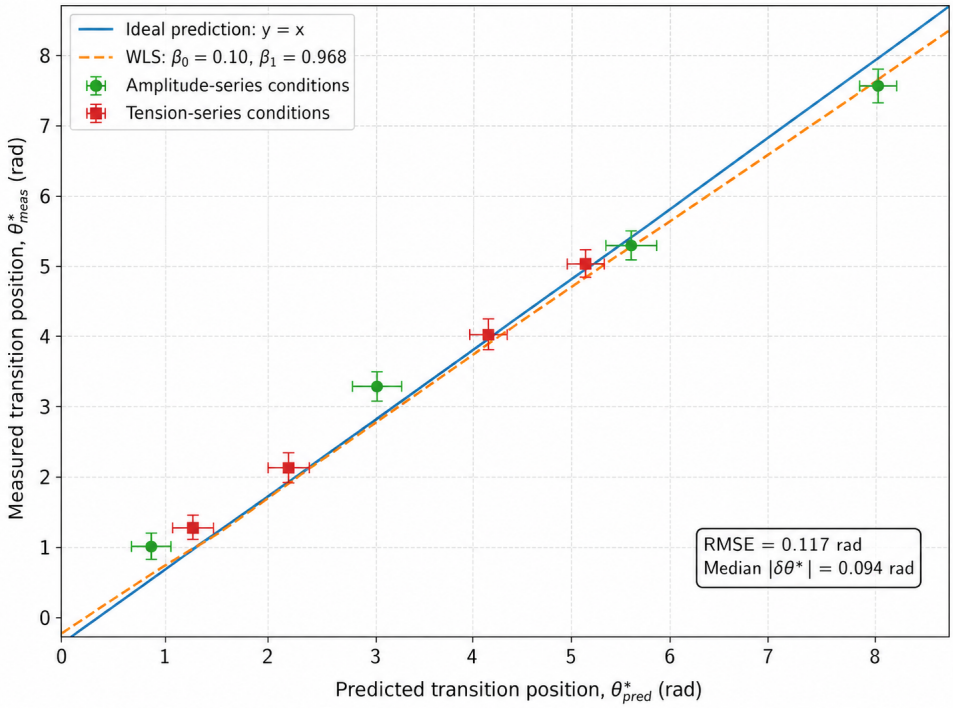


Figure 9: Measured versus independently predicted transition positions for the eight unique stable-front conditions. Horizontal and vertical error bars denote prediction and measurement uncertainty, respectively. The solid line is the identity relation, and the dashed line is the weighted experimental regression.

The calibration-fixed law therefore reproduced the absolute transition position across the tested stable-front conditions without fitting to the measured front locations. Both series are anchored at the reference condition, whose predicted position is π by construction because $u_{0,\text{ref}}$ was obtained by inverting the same law, so β_0 and β_1 largely re-express the two slope comparisons already reported. The unanchored absolute result is the reference condition itself, for which $\theta^*_{\text{meas}} = 3.27 \pm 0.17$ rad against the predicted $\pi = 3.14$ rad, a difference of 0.13 rad. A second unanchored comparison is the onset threshold: the calibrated $u_{0,\text{th}} = 0.1875$ mm lies between the highest amplitude that produced no accepted front, 0.1612 mm, and the lowest that did, 0.2047 mm. Neither test uses the fitted transition positions to set any model parameter. (Fig. 9)

4.6 Frequency Falsification Test

At fixed measured displacement amplitude and boundary loading, the excitation frequency was varied over

$$f = \{30, 40, 50, 60\} \text{ Hz}. \tag{92}$$

All four conditions produced stable creep-grip transitions at

$$\theta^*_{\text{meas}} = 3.27 \pm 0.17,\ 3.15 \pm 0.18,\ 3.36 \pm 0.20,\ 3.21 \pm 0.18 \text{ rad}, \tag{93}$$

for $f = 30$, 40, 50, 60 Hz, respectively. No monotonic shift in transition position was observed over the tested frequency range. The reduced model contains no explicit frequency dependence when the measured displacement amplitude is held fixed. This was tested using

$$\theta^* = C_f + \beta_f \ln\left(\frac{f}{f_{\text{ref}}}\right), \qquad f_{\text{ref}} = 30 \text{ Hz}, \tag{94}$$

for which the reduced-model prediction is

$$\beta_{f,\text{theory}} = 0. \tag{95}$$

Weighted least-squares regression of the four measured transition positions gave

$$\boxed{\beta_{f,\text{meas}} = -0.003 \pm 0.385 \text{ rad}} \tag{96}$$

at the 95% confidence level (Fig. 10). The fitted slope was therefore consistent with zero within uncertainty.

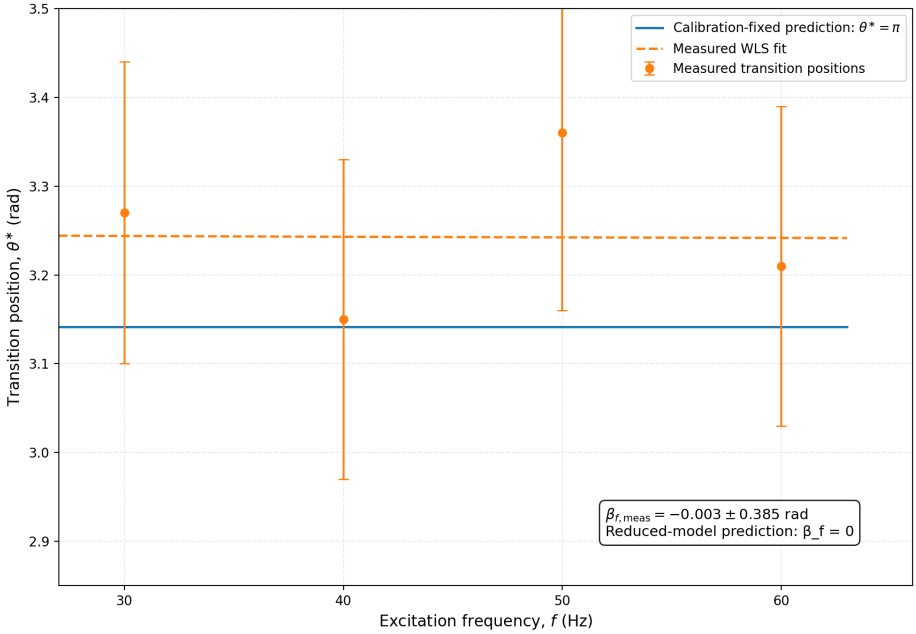


Figure 10: Frequency falsification test at fixed measured displacement amplitude and boundary loading. The solid line is the calibration-fixed reference prediction $\theta^* = \pi$, while the dashed line is the weighted experimental fit. Error bars denote the 95% uncertainty in the measured transition positions.

No systematic frequency dependence was resolved over 30–60 Hz, consistent with the reduced model over the tested range.

4.7 Uncertainty and Robustness of the Measured Transition

The robustness criteria defined in Section 3.6.2 were applied to the eleven of the fourteen unique operating conditions that produced accepted stable fronts. The largest observed instability for each diagnostic remained within its predefined acceptance limit.

The largest leave-one-repeat and leave-one-position shifts were 0.12 rad and 0.08 rad, respectively, while the maximum repeat-centre dispersion was 0.074 rad. The highest classification flip fraction was 2.0%, below the predefined 5% limit.

Neither of the two control conditions produced an accepted stable transition across its six experimental repeats. Across the eight unique conditions used for the absolute-prediction test, the largest absolute residual was 0.186 rad, while the overall root-mean-square prediction error was

$$\mathrm{RMSE} = 0.117 \text{ rad}. \tag{97}$$

The robustness results are summarized in Table 3.

Table 3: Largest observed instability across the eleven accepted stable-front conditions compared with the predefined acceptance limits.

Diagnostic	Observed value	Acceptance limit
Bootstrap SD of θ^*	0.09 rad	≤ 0.14 rad
Relative bootstrap SD of w	0.24	≤ 0.35
Repeat-centre dispersion	0.074 rad	≤ 0.105 rad
Maximum leave-one-position shift	0.08 rad	≤ 0.14 rad
Maximum leave-one-repeat shift	0.12 rad	≤ 0.18 rad
Classification-flip fraction	2.0%	$\leq 5\%$
Control conditions with false fronts	0/2 conditions (0/12 runs)	0 accepted

The accepted values of θ^* were therefore robust to repeat removal, spatial deletion, and resampling, with no false stable-front classifications in the control runs.

4.8 Condition-Level Experimental Summary

The fourteen unique operating conditions that underlie the 84 physical runs are reported in Appendix A. Appendix A reports the actual values of each of the parameters u_0, T_1, T_2, and f that were used to define each of the operating conditions, as well as the classification of each condition as containing a stable front or not, and provides the independently predicted and measured values of the transition position θ^* and the fitted transition width w. Each operating condition was prepared six times as a physical experiment. Those experiments that did not exhibit stable fronts were retained as no-accepted-front experimental outcomes and do not have any value assigned to θ^* or w.

5.0 Discussion

The results of the experiments support the basic physical picture that is proposed in this work. Each of the experiments that produced an accepted stable front produced a front that was stable within the spatial portion of the contact that was wrapped around the cylinder. In each case, the rate of creep was the highest near the held end, decreased over a finite portion of the wrap to the rate at which the gripping contact was expected to occur at any point along the wrap, and the front between those two rates had the characteristics described above. The fact that experiments that used no vibration at all did not exhibit stable fronts indicates that the existence of a stable front was not an artifact of the fitting of the video sequences to determine the positions of the stable front. Instead, the direction of movement of the stable front was in the expected direction. Increasing the displacement amplitude of the vibration moved the stable front toward the loaded end of the contact; and increasing the tension at the held end moved the stable front toward the held end of the contact.

The amplitude series gave

$$\beta_{a,\mathrm{meas}} = 9.54 \pm 0.74 \text{ rad}, \qquad \beta_{a,\mathrm{theory}} = 10.10 \pm 1.45 \text{ rad},$$

while the tension series gave

$$\beta_{T,\mathrm{meas}} = -5.56 \pm 0.50 \text{ rad}, \qquad \beta_{T,\mathrm{theory}} = -5.56 \pm 0.62 \text{ rad}.$$

Read together, the two slopes recover the calibrated parameters rather than merely agreeing with them. Since $\beta_T = -1/\mu_{\mathrm{red}}$ and $\beta_a = 1/[(1-\alpha)\mu_{\mathrm{red}}]$, the tension slope alone gives $\mu_{\mathrm{red}} = 0.180 \pm 0.016$, against the independently calibrated 0.18 ± 0.02; and the ratio of the two slopes, $1 - |\beta_T|/\beta_a$, gives $\alpha = 0.417 \pm 0.069$, against the independently calibrated 0.45 ± 0.05. The second of these is notable because it uses no friction calibration at all: the stiffness exponent measured in the preslip calibration and the exponent implied by the two transition-position slopes agree within their propagated uncertainties. The apparent shortfall of $\beta_{a,\mathrm{meas}}$ relative to $\beta_{a,\mathrm{theory}}$ is the same difference expressed in the other direction.

The tension dependence is especially informative about the dependence of β_T on the stiffness exponent α; its value of $-1/\mu_{\mathrm{red}}$ is independent of α. For the eight unique conditions of the amplitude and tension series, the model, using the calibrated variables μ_s, μ_{red}, κ_{ref}, and α, together with the measured u_0 and T_1 for each condition, gave:

$$\beta_0 = 0.10 \pm 0.13 \text{ rad}, \qquad \beta_1 = 0.968 \pm 0.035,$$

relative to the ideal values $\beta_0 = 0$ and $\beta_1 = 1$; the RMSE of the predictions is 0.117 rad, and the median absolute error is 0.094 rad. These values are obtained without using the measured positions of the transitions to calibrate the model. An additional test of the model uses the frequency-series measurements at fixed displacement amplitude and boundary loading:

$$\beta_{f,\mathrm{meas}} = -0.003 \pm 0.385 \text{ rad},$$

which was also consistent with zero over the frequency range of 30 to 60 Hz. Thus, within this frequency range, the position of the front was better described as being controlled by the displacement amplitude than by the frequency of the applied motion. This does not exclude the influence of rate-dependent friction, dynamic stiffening, inertia, or other effects outside of the frequency range used in these experiments.

Two of the calibrated features of the model are the values of the static and reduced friction coefficients. The condition that the static coefficient μ_s is greater than the reduced coefficient μ_{red} permits treating the breakaway of the front from the rest of the wrap as a separate process from the determination of the effective resistance state within the wrap. The second is a measured law for the stiffness of the wrap as a function of tension:

$$\kappa_t(T) = \kappa_{\mathrm{ref}} \left(\frac{T}{T_{\mathrm{ref}}} \right)^{\alpha}$$

with $\alpha < 1$ ensures that vibration-induced loading grows with wrap angle more slowly than the effective frictional resistance, allowing an interior crossover to exist. The finite observed transition width is consistent with the distributed compliance and heterogeneous contact of a braided rope rather than an ideal discontinuity.

The model is thus only a model of the transition centre, not of the entire rope–capstan interface. The model

does not consider the local distributions of pressure along the capstan, the individual contacts between the rope's fibres and the capstan, the internal forces that are transmitted along the rope's fibres during one cycle of oscillation, the slippage of the fibres within the braid, or the detailed evolution of the transition width along the rope. Likewise, the concept of $T_{\mathrm{eff}}(\theta)$ does not refer to the instantaneous physical tension of the rope at any given phase of its oscillation. Instead, the model is only applicable to the regime of partial mobilisation in which a single stable creep-grip front was observed; below the threshold of mobilisation no stable front was observed, and any migrating or multi-patch fronts exist outside the assumptions of the single-front law.

Within the accepted regime, however, the position of the transition zones was robust to bootstrap resampling, repeat deletion, and spatial deletion, and none of the twelve control experiments produced a false stable front. Thus, the study extends the capstan picture to determine where local mobilisation of a rope upon a cylindrical surface can occur; the gradient of frictional resistance along the wrap can determine the position of the interior boundary between the creeping and gripping portions of the rope.

Consequently, the main result of the study is that the position of the transition between creeping and gripping of a vibratory rope can be predicted from independently measured parameters of the rope, the capstan, the vibratory motion, and the tension of the rope's held end. Thus, the model can be used for the analysis and potential control of the partial mobilisation of a vibratory rope.

6.0 Evaluation and Error Analysis

The uncertainties in the predicted value of θ^* are dominated by those in μ_{red}, α, and κ_{ref} because of their propagated sensitivities and calibration uncertainties. The contributions of the uncertainty in the remaining parameters (r, a, T_1, and μ_s) are smaller. Rather than linearising the transition law, the uncertainties in the individual calibrated parameters were propagated through the law to determine the uncertainty in θ^*.

The uncertainty in the measured value of θ^* was dominated by the uncertainty in the optical resolution of the measurements of the position of the transition front and repeat-to-repeat variation in the fitted front location. The uncertainty in the registration of the direct and reflected views of the contact was accounted for within the uncertainty in the angular parameter, and reduced by fixing the registration of the views prior to confirmatory testing.

Finally, the measured value of the transition angle was not based on measuring the angle of a discontinuity in the contact. Thus, the uncertainty in θ^* also reflects the finite fitted width of the transition region w. This width is physically expected from the compliance of the braided rope, the heterogeneous nature of the local contact, and the distributed preslip deformation. The use of a common centre and width value across the six repeats has allowed the model to be less sensitive to local fluctuations in geometry while preserving genuine differences in amplitude and offset between the repeats.

A second potential source of uncertainty relates to the identifiability of the model. More specifically, a broad distributed creep component along the contact can exhibit the same features as a very wide front if the angular component of the front is poorly resolved. Thus, alternative response models were developed and used to evaluate the same contact profiles; a value of θ^* was assigned to each condition only if the stable-front model satisfied the predefined BIC, signal, spatial-resolution, migration, and resampling criteria. Profiles that could not be distinguished from the alternative models were retained as non-front or unresolved outcomes and were not included in the predictive comparison.

The conversion from finite-contact tangential stiffness K_t to stiffness per unit wrap angle,

$$\kappa_t = \frac{K_t}{\Delta\theta}, \tag{98}$$

introduces a model assumption that the preslip stiffness is approximately extensive over the short calibration contact. The additional calibrations with short rope contact lengths support the assumption of an approximately linear dependence of the factor K_t on the quantity $\Delta\theta$ over the range of contact angles employed here. However, the assumption has not been established for arbitrary contact lengths and materials. Consequently, the extrapolation of this stiffness law to different rope geometries should therefore be made cautiously.

A related limitation is the use of

$$T_{\text{eff}}(\theta) = T_1 e^{\mu_{\text{red}}\theta} \tag{99}$$

as an effective resistance-state profile for the rope. This resistance state is not the rope's tension at every phase of the vibrational cycle. The model does not consider transient effects associated with the vibration cycle, the distribution of normal pressure along the rope's length, the contact of individual rope fibres, or the microslips that may occur within the braid itself. The model's purpose is limited to the prediction of the centre of a stable partial-mobilisation front.

A separate limitation arises from the loading design. Because a fixed ratio $T_2/T_1 = 5.00$ was held throughout the campaign, the held-end and loaded-end tensions were varied together and are perfectly collinear across the tension series. The measured β_T therefore cannot distinguish a dependence on T_1 alone from a dependence on the tension pair at fixed ratio. The reduced model attributes the crossover position to the held-end resistance scale T_1, with T_2 entering only through the admissibility window of Eq. 40. Testing that attribution requires a loaded-end sweep at fixed T_1 and fixed u_0, which lies outside the present campaign. The values reported here should therefore be read as applying to the fixed-ratio loading family examined.

The frequency result also defines a finite range of validity for the established result. While no frequency dependence was resolved between 30 and 60 Hz at fixed measured displacement amplitude, this only establishes that the result is valid within the tested frequency range. At higher frequencies, or for different materials, other factors that contribute to the motion of the front may be more significant than the factors considered in this study. Thus, the conclusions of this study are limited to the tested material pair, four-wrap geometry, loading and displacement ranges, and the partial-mobilisation regime. Other loading conditions outside of those considered in this study (with migrating fronts, multiple localized slip regions, no resolved front, etc.) are outside of the domain of the single-transition law, and were, therefore, not included in the test of the predictions of the law.

Within the limits of the factors considered, the results of the acceptance tests indicate the robustness of the model. The positions of the transitions were stable during the repeat and spatial resampling tests, no stable front was accepted in the twelve control runs, and the comparison of the measured and predicted values was close to the identity relation within the accepted operating conditions of the system (RMSE = 0.117 rad). Thus, the uncertainties inherent to the model constrain the range and precision of the model, but do not alter the central conclusion regarding the ability to predict the position of the stable front.

7.0 Conclusion

The present work investigates the occurrence of a creep-grip transition along a rope due to torsional vibration of the capstan and the predictability of the position of that transition based upon independently measurable rope and capstan quantities. The results indicate that, above the onset threshold, cycle-accumulated relative motion between the rope and capstan develops near the end of the rope that is held, while the loaded-end region remains in the gripping state within measurement resolution. The resulting transition is finite in width and has a centre that can be denoted as θ^*. Neither the control experiment with no vibrational input into the capstan, nor the experiment with vibrational input that is below the threshold value for initiating a creep-grip transition, produces an accepted stable front. The position of the transition front is found to vary with both the amplitude of the vibrational input into the capstan, as well as with the tension at the held end of the rope. The series of experiments with varying amplitudes of vibrational input into the capstan resulted in

$$\beta_{a,\mathrm{meas}} = 9.54 \pm 0.74 \text{ rad}, \tag{100}$$

in agreement with the independently predicted

$$\beta_{a,\mathrm{theory}} = 10.10 \pm 1.45 \text{ rad}. \tag{101}$$

Similarly, the tension series gave

$$\beta_{T,\mathrm{meas}} = -5.56 \pm 0.50 \text{ rad}, \tag{102}$$

compared with the prediction

$$\beta_{T,\mathrm{theory}} = -5.56 \pm 0.62 \text{ rad}. \tag{103}$$

The absolute prediction of the transition position was also tested. Using only independently calibrated values of μ_s, μ_{red}, κ_{ref}, α, u_0, and T_1, the predictive law gave measured-versus-predicted agreement characterised by

$$\beta_0 = 0.10 \pm 0.13 \text{ rad}, \qquad \beta_1 = 0.968 \pm 0.035, \tag{104}$$

relative to the ideal values $\beta_0 = 0$ and $\beta_1 = 1$. The corresponding root-mean-square prediction error was 0.117 rad and the median absolute angular error was 0.094 rad.

At fixed measured displacement amplitude, no systematic frequency dependence was resolved over 30–60 Hz, with

$$\beta_{f,\mathrm{meas}} = -0.003 \pm 0.385 \text{ rad}, \tag{105}$$

consistent with the displacement-controlled reduced model over the tested range.

The model, then, describes only the centre of a stable partial-mobilisation front—it is not a model of the dynamics of the entire capstan–rope interface. Within that regime, though, the position of the transition from vibration-induced creep to grip is found to be non-arbitrary and predictable from independently calibrated

friction, contact stiffness, geometry, loading, and vibration amplitude. The main result of this study is, therefore, a closed-form expression for that relationship—a scaling law for the position of the vibration-induced transition between creep and grip in capstan friction—which is experimentally supported over the regime of partial mobilisation that was studied.

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A Experimental Operating Conditions and Transition Measurements

The confirmatory campaign contained fourteen unique operating conditions. Each condition was repeated six times using an independently prepared rope–capstan contact region, giving

$$14 \times 6 = 84 \tag{106}$$

physical runs. The 30 Hz reference condition was shared between the amplitude, held-end-tension, and frequency series and is therefore listed only once.

The reference operating values were

$$T_{1,\mathrm{ref}} = 1.00\ \mathrm{N}, \qquad u_{0,\mathrm{ref}} = 0.2559\ \mathrm{mm}, \qquad \frac{T_2}{T_1} = 5.00. \tag{107}$$

Using the independently calibrated values $\mu_s = 0.30$ and $\kappa_{\mathrm{ref}} = 1.60\ \mathrm{kN\,m^{-1}\,rad^{-1}}$, the onset displacement at the reference tension was

$$u_{0,\mathrm{th}} = \frac{\mu_s T_{1,\mathrm{ref}}}{\kappa_{\mathrm{ref}}} = \boxed{0.1875\ \mathrm{mm}}. \tag{108}$$

The sub-threshold control was therefore

$$u_{0,\mathrm{control}} = 0.50 u_{0,\mathrm{th}} = 0.09375\ \mathrm{mm}. \tag{109}$$

No value of θ^* or transition width w was assigned when the predefined stable-front criteria were not satisfied.

Table 4: Complete condition-level summary of the fourteen unique operating conditions underlying the 84 physical runs. Each condition contained six independently prepared repeats. Predicted transition positions were calculated using the independently fixed values $\alpha = 0.45$ and $\mu_{\mathrm{red}} = 0.18$. A negative formal prediction lies outside the physical domain and therefore denotes a predicted no-front condition. No value of θ^* or w was assigned when no stable front was accepted.

Condition	u_0 (mm)	T_1 (N)	T_2 (N)	f (Hz)	T_2/T_1	Observed class	θ^*_{pred} (rad)	θ^*_{meas} (rad)	w (rad)
A0 – Amplitude	0.1612	1.00	5.00	30	5.00	No accepted front	-1.53	—	—
A1 – Amplitude	0.2047	1.00	5.00	30	5.00	C_{stable}	0.89	1.03 ± 0.17	0.46
R – Reference	0.2559	1.00	5.00	30	5.00	C_{stable}	3.14	3.27 ± 0.17	0.52
A2 – Amplitude	0.3224	1.00	5.00	30	5.00	C_{stable}	5.48	5.29 ± 0.18	0.57
A3 – Amplitude	0.4043	1.00	5.00	30	5.00	C_{stable}	7.76	7.60 ± 0.22	0.61
T0 – Tension	0.2559	0.71	3.55	30	5.00	C_{stable}	5.04	5.10 ± 0.15	0.55
T1 – Tension	0.2559	0.84	4.20	30	5.00	C_{stable}	4.11	4.05 ± 0.13	0.49
T2 – Tension	0.2559	1.19	5.95	30	5.00	C_{stable}	2.18	2.12 ± 0.12	0.53
T3 – Tension	0.2559	1.41	7.05	30	5.00	C_{stable}	1.23	1.28 ± 0.14	0.47
C0 – No vibration	0	1.00	5.00	—	5.00	No accepted front	—	—	—
C1 – Sub-threshold	0.0938	1.00	5.00	30	5.00	No accepted front	-7.00	—	—
F40 – Frequency	0.2559	1.00	5.00	40	5.00	C_{stable}	3.14	3.15 ± 0.18	0.50
F50 – Frequency	0.2559	1.00	5.00	50	5.00	C_{stable}	3.14	3.36 ± 0.20	0.59
F60 – Frequency	0.2559	1.00	5.00	60	5.00	C_{stable}	3.14	3.21 ± 0.18	0.54

The reference condition R is shared by the amplitude, held-end-tension, and frequency series and is therefore listed only once. Together with the other eight primary conditions, two controls, and three additional frequency conditions, the table represents fourteen unique conditions and $14 \times 6 = 84$ physical runs.

Eleven of the fourteen conditions produced accepted stable transitions. The three non-front conditions were the no-vibration control, the sub-threshold control, and the lowest-amplitude primary condition.

Across the eight unique amplitude–tension operating points used in the absolute prediction test, the condition-wise residuals were

$$\delta\theta^* = \{+0.142,\ +0.128,\ -0.186,\ -0.162,\ +0.056,\ -0.060,\ -0.055,\ +0.047\}\ \text{rad}, \tag{110}$$

giving

$$\boxed{\mathrm{RMSE} = 0.117\ \text{rad}}, \qquad \boxed{\text{median}\,|\delta\theta^*| = 0.094\ \text{rad}}. \tag{111}$$

The five tension-series residuals change sign across the operating range without a systematic offset. The four amplitude-series residuals, ordered by increasing u_0, decrease monotonically from $+0.142$ to -0.162 rad. This is a slope deficiency rather than an offset, and corresponds to the small difference between the measured and independently calibrated stiffness exponent discussed in Section 5.0.

B Experimental Parameters, Calibration, and Reproducibility

The quantities entering the transition-position law were calibrated independently of the measured transition positions. No measured value of θ^* was used to determine r, μ_s, μ_{red}, κ_{ref}, or α.

The calibrated stiffness relation was

Table 5: Principal apparatus, calibration, and acquisition quantities used in the experiment.

Quantity	Value / specification	Role
Capstan	6061 aluminium, 25.00 ± 0.06 mm diameter	$r = 12.50 \pm 0.03$ mm
Rope	3.0 mm braided polyester	Fresh four-wrap region per run
Total wrap	$\Theta = 8\pi = 25.133$ rad	Four complete turns
Static friction	$\mu_s = 0.30 \pm 0.02$	Quasi-static breakaway
Reduced friction	$\mu_{\mathrm{red}} = 0.18 \pm 0.02$	Slow-creep calibration
Tangential stiffness	$\kappa_{\mathrm{ref}} = 1.60 \pm 0.12$ kN m^{-1} rad^{-1}	Preslip calibration
Stiffness exponent	$\alpha = 0.45 \pm 0.05$	$\kappa_t \propto T^{\alpha}$
Calibration contact	$\Delta\theta = \pi/6 = 30^{\circ}$	Preslip stiffness measurement
Reference tension	$T_{\mathrm{ref}} = 1.00$ N	Stiffness reference
Loading ratio	$T_2/T_1 = 5.00$	Primary and frequency tests
High-speed imaging	1000 frames s^{-1}, ≈ 0.20 ms exposure	Rope/capstan tracking
Optical displacement floor	$\sigma_{\mathrm{opt}} = 0.015 \pm 0.003$ mm	Tracking resolution
Creep-rate floor	$\sigma_v = 0.10 \pm 0.02$ μm cycle^{-1}	Grip measurement floor
Signal threshold	$3\sigma_v = 0.30$ μm cycle^{-1}	Front acceptance criterion
Spatial representation	120 bins	12° nominal spacing
Phase-matched samples	$N = 0, 5, \ldots, 150$	31 cycle samples per position
Physical repeats	6 per condition	Independent contact regions

$$\boxed{\kappa_t(T) = 1.60 \left(\frac{T}{1.00\ \mathrm{N}} \right)^{0.45} \ \mathrm{kN\,m^{-1}\,rad^{-1}}}. \tag{112}$$

Evaluated at the five held-end tensions used in the confirmatory tension series, this gives

T (N)	0.71	0.84	1.00	1.19	1.41
κ_t (kN m^{-1} rad^{-1})	1.37	1.48	1.60	1.73	1.87

The chronological protocol for every physical run was

$$\boxed{\begin{array}{c} 60\ \mathrm{s\ settling} \rightarrow 5\ \mathrm{s}\ T_1\ \mathrm{pre\text{-}measurement} \rightarrow 1\ \mathrm{s\ baseline} \\ \rightarrow 2\ \mathrm{s\ spin\text{-}up} \rightarrow 5\ \mathrm{s\ high\text{-}speed\ recording} \rightarrow 5\ \mathrm{s}\ T_1\ \mathrm{post\text{-}measurement} \\ \rightarrow \mathrm{fresh\ rope\text{-}contact\ region}. \end{array}} \tag{113}$$

For vibration-on runs, a run was retained only when

$$\frac{|T_{1,\mathrm{post}} - T_{1,\mathrm{pre}}|}{T_{1,\mathrm{pre}}} \leq 2\%, \qquad \frac{|\Delta u_0|}{u_0} \leq 3\%. \tag{114}$$

For the no-vibration control, the displacement-amplitude criterion was not applicable because the vibration drive remained disabled.

At 30 Hz, the 5 s recording contained approximately 150 cycles. The corresponding recordings at 40, 50, and 60 Hz contained approximately 200, 250, and 300 cycles, respectively. For direct comparison between frequencies, the analysis used the common interval from cycle 0 to cycle 150.

The complete optical processing sequence was

$$\begin{array}{c} \mathrm{video} \rightarrow \mathrm{capstan/rope\ tracking} \rightarrow \mathrm{view\ registration} \rightarrow \mathrm{phase\ matching} \\ \rightarrow u_{\mathrm{rel}}(\theta, N) \rightarrow v_c(\theta) \rightarrow \mathrm{candidate\text{-}model\ fitting} \rightarrow \theta^*,\ w \text{ or no accepted front}. \end{array} \tag{115}$$

C Transition Classification and Robustness Diagnostics

Each experimental condition was analysed using the same competing spatial models described in Section 3.6: no resolved structure (A), distributed creep (B), a stable localized transition (C_{stable}), a repeat-dependent migrating transition ($C_{\mathrm{migrating}}$), and multiple localized regions (D).

Bayesian Information Criterion was evaluated as

$$\mathrm{BIC} = n \ln\left(\frac{\mathrm{RSS}}{n}\right) + k \ln n. \tag{116}$$

A stable front was accepted only when all criteria in Table 6 were satisfied.

Table 6: Predefined stable-transition acceptance criteria and worst observed values across the eleven accepted C_{stable} conditions.

Diagnostic	Acceptance limit	Worst observed
Stable model over non-front alternatives	$\Delta\mathrm{BIC} \geq 10$	Passed accepted conditions
Front amplitude	$> 3\sigma_v = 0.30\ \mu\mathrm{m\,cycle}^{-1}$	Passed accepted conditions
Centre-to-boundary clearance	≥ 0.35 rad	Passed accepted conditions
Migrating-model improvement	< 10 BIC units	Passed accepted conditions
Repeat-centre dispersion	≤ 0.105 rad	0.074 rad
Bootstrap SD of θ^*	≤ 0.14 rad	0.09 rad
Relative bootstrap SD of w	≤ 0.35	0.24
Leave-one-position shift	≤ 0.14 rad	0.08 rad
Leave-one-repeat shift	≤ 0.18 rad	0.12 rad
Classification-flip fraction	$\leq 5\%$	2.0%
False fronts in controls	0 accepted	0/12 runs

For the representative reference condition,

$$u_0 = 0.2559\ \mathrm{mm}, \qquad T_1 = 1.00\ \mathrm{N}, \qquad f = 30\ \mathrm{Hz}, \tag{117}$$

the stable model gave

$$\theta^*_{\mathrm{meas}} = 3.27 \pm 0.17\ \mathrm{rad}, \qquad w = 0.52\ \mathrm{rad}. \tag{118}$$

Its BIC advantages over the non-front alternatives were

$$\Delta\mathrm{BIC}_A = 31.8, \qquad \Delta\mathrm{BIC}_B = 19.6, \qquad \Delta\mathrm{BIC}_D = 16.9. \tag{119}$$

Allowing repeat-dependent front positions improved the BIC by only

$$5.2 < 10, \tag{120}$$

so the additional migrating-front parameters were not supported.

Across the eleven accepted stable-front conditions, the fitted widths lay in

$$0.46 \leq w \leq 0.61\ \mathrm{rad}, \tag{121}$$

with mean width

$$\overline{w} \approx 0.53 \text{ rad.} \tag{122}$$

The finite width and non-zero repeat/deletion variations indicate measurable experimental scatter rather than an artificially exact transition. At the same time, all accepted conditions remained inside the predefined robustness limits, and neither of the two control conditions produced an accepted stable front.